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**Hourly Multi-Pollutant Air Quality Forecasting using BiLSTM-GRU Based Deep Learning
Model on Chandni Chowk and R.K. Puram, Delhi**

submitted for partial fulfilment of the requirements for the award of the degree of

BACHELOR IN COMPUTER APPLICATIONS [B.C.A]

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GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY, NEW DELHI

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DECLARATION BY THE CANDIDATE

I hereby declare that the work, which is being presented in this project entitled **“Hourly Multi-Pollutant Air Quality Forecasting using BiLSTM-GRU Based Deep Learning Model on Chandni Chowk and R.K. Puram, Delhi”**, is an authentic record of my own work carried out by me under the supervision and guidance of **“DR. NAUSHAD AHMAD”** project guide.

This project was undertaken as a part of the major project report as per the curriculum of GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY, DELHI for the partial fulfilment of B.C.A from KCC INSTITUTE OF LEGAL AND HIGHER EDUCATION.

The matter embodied in this project work has not been submitted earlier for award of any degree or diploma to the best of my knowledge and belief.

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CERTIFICATE FROM GUIDE

This is to certify that project title “**Hourly Multi-Pollutant Air Quality Forecasting using BiLSTM-GRU Based Deep Learning Model on Chandni Chowk and R.K. Puram, Delhi**” is the original work of “NIHIRA DUBEY” Enroll No.- 00927402023, Student of BACHELOR OF COMPUTER APPLICATIONS, 6th Semester, has duly completed her project under my guidance and supervision up to my satisfactory level.

This work has been done in partial fulfilment of the requirement for the award of the degree of Bachelor of Computer Applications from **KCC Institute of Legal and Higher Education** through **Guru Gobind Singh Indraprastha University**, New Delhi and has not been submitted anywhere in any other university for the award of any degree.

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On the basis of Project Report submitted by “**NIHIRA DUBEY**”, student of Bachelor of Computer Applications, I hereby certify that the Project Report “**HOURLY MULTI-POLLUTANT AIR QUALITY FORECASTING USING BiLSTM-GRU BASED DEEP LEARNING MODEL ON CHANDNI CHOWK AND R.K. PURAM, DELHI**” Which is submitted to the Department of Computer Applications, KCC Institute of Legal and Higher Education, in partial fulfilment of requirement for the award of the degree of BACHELOR OF COMPUTER APPLICATIONS is an original contribution with existing knowledge and faithful record of work carried out by him/her under my guidance and supervision. To the best of my knowledge this work has not been submitted in part or full for any Degree or Diploma to this University or elsewhere.

DATE: - 13-04-2026

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ACKNOWLEDGEMENT

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CHAPTER 1- INTRODUCTION

Hourly Multi-Pollutant Air Quality Forecasting using BiLSTM-GRU Based Deep Learning Model on Chandni Chowk and R.K. Puram, Delhi

1 INTRODUCTION

Air pollution has become a serious environmental and public health issue worldwide. This is mainly due to rapid industrial growth, urban expansion, and increased vehicle emissions. Harmful pollutants like particulate matter ($PM_{2.5}$ and PM_{10}), carbon monoxide (CO), nitrogen dioxide (NO_2), and ozone (O_3) can greatly impact human health and ecosystems [1]. Poor air quality leads to respiratory diseases, heart problems, and higher death rates, making it a major global challenge [2]. Recently, the need for effective monitoring and predicting of air quality has gained attention. Governments and environmental agencies set up monitoring stations to gather real-time data; however, these stations are often few in number and cannot cover all areas. This creates a need for smart systems that can analyze past data and accurately predict future air quality levels.

Advances in machine learning and deep learning techniques have created new possibilities for air quality forecasting [3]. Traditional statistical models often struggle to capture the complex relationships found in environmental data. In contrast, deep learning models, such as Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Bidirectional Long Short-Term Memory (Bi-LSTM), along with hybrid models, have shown promising results in time-series predictions [4]. These models can identify patterns from large datasets and deliver more accurate forecasts. This study looks at the comparative analysis of different traditional and hybrid deep learning models for predicting air quality [5]. By assessing models using several performance metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2), the research aims to find the most effective way to predict air quality levels.

Air quality assessment relies on analyzing various atmospheric pollutants and weather factors that affect environmental conditions and human health [6]. Key pollutants include particulate matter, such as $PM_{2.5}$ and PM_{10} . $PM_{2.5}$ consists of particles smaller than 2.5 micrometers [7]. These tiny particles can penetrate deep into the lungs and even enter the bloodstream, leading to serious respiratory and heart diseases. PM_{10} comprises larger particles up to 10 micrometers in diameter, which mainly impact the upper respiratory system and can cause irritation and breathing issues. These particles come from sources like vehicle emissions, industrial processes, construction work, and natural events such as dust storms. Nitrogen dioxide (NO_2)

is another major air pollutant released primarily from combustion processes, especially from vehicles, power plants, and industrial activities. It contributes to the formation of smog and is linked to respiratory diseases like asthma and bronchitis. Carbon monoxide (CO) is a colorless, odorless gas generated by incomplete burning of carbon-based fuels. It is highly toxic because it decreases the oxygen-carrying capacity of hemoglobin in the blood, posing serious health risks, especially in high concentrations. Ground-level ozone (O₃) is a secondary pollutant formed through complex chemical reactions involving nitrogen oxides and volatile organic compounds in sunlight [8]. Unlike stratospheric ozone, which protects the Earth from harmful UV rays, ground-level ozone is harmful and can irritate the respiratory system, reduce lung function, and damage plants. Ozone levels depend heavily on environmental factors like sunlight and temperature, making it a complicated pollutant to study. Beyond these pollutants, weather conditions play a vital role in how pollutants disperse, transform, and accumulate in the atmosphere. It is crucial for forming secondary pollutants like ozone and it affects temperature changes, which in turn influence pollutant behavior and chemical reactions [9].

The interaction of different pollutants and weather factors creates a complex system that shapes overall air quality. These interactions can be non-linear and change over time, making analyzing and predicting pollution levels difficult with traditional methods. For example, high solar radiation combined with nitrogen oxides can significantly boost ozone levels, while high humidity and low wind speed can lead to particulate matter buildup. Thus, it is important to consider both pollutant levels and weather conditions together for an accurate air quality assessment. In modern studies of air quality, combining multi-pollutant data with weather factors has become essential for creating reliable predictive models. Advanced techniques like machine learning and deep learning help analyze these complex relationships and forecast future air quality. By including parameters such as PM_{2.5}, PM₁₀, NO₂, CO, O₃, these models can capture changes in air pollution over time and space [10]. This thorough approach improves the accuracy of predictions and aids effective environmental management and policymaking. Overall, analyzing air pollutants along with weather factors provides a complete understanding of air quality dynamics. It helps researchers and policymakers identify pollution sources, evaluate health risks, and enforce suitable control measures. As air pollution remains a significant global challenge, studying these parameters is crucial for enhancing air quality monitoring systems and safeguarding public health [11].

Solar radiation drives reactions in the air and is important for forming secondary pollutants like ozone [12]. It also affects temperature changes, which influence how bad things in the air behave and react. The interaction of bad things in the air and weather conditions creates a complex system that shapes the overall air quality. In order to understand air quality it is very important to look at both things in the air and weather conditions together. Advanced techniques like machine learning and deep learning can help analyze these relationships and predict future air quality. By including parameters like PM_{2.5}, PM₁₀, NO₂, CO, O₃ these models can capture

changes in air pollution over time and space. Looking at bad things in the air at the same time is a useful approach in understanding air quality. It helps us better understand how these bad things affect the environment and human health. In the past studies on air pollution often looked at one thing at a time. However in life people are exposed to many bad things in the air that interact with each other in complex ways [13]. The main bad things in the air that are looked at together include particles, nitrogen dioxide, sulfur dioxide, carbon monoxide and ozone. These bad things come from sources like vehicles, factories, power generation and burning fossil fuels. Their levels in the air often relate to each other because they undergo changes in the air. A major aspect of looking at bad things in the air at the same time is how they interact with each other. For example nitrogen oxides and other things react with sunlight to form ground-level ozone, which's a harmful secondary pollutant. Similarly nitrogen dioxide can lead to the formation of tiny particles through reactions in the air.

These interactions show that it is necessary to study things in the air together instead of one at a time. Focusing on one thing may ignore important chemical links and combined health effects. Research shows that being exposed to bad things in the air can have worse health effects than being exposed to one bad thing. Looking at bad things in the air at the same time is also vital for predicting air quality and making models. Air pollution data is complex because it includes variables that change over time and are influenced by factors like weather conditions. Advanced techniques like machine learning and deep learning models are increasingly used to analyze data on bad things in the air. These models can capture -linear relationships among bad things in the air leading to more precise predictions of air quality levels [14]. By looking at bad things in the air at once these models can better reflect real-world situations and improve forecasting performance. Looking at bad things in the air at the same time is important for environmental policy and regulation. Governments and environmental agencies use air quality indices that often rely on bad things in the air to evaluate overall air quality and issue health warnings. Understanding how bad things in the air behave together helps policymakers create strategies to control pollution. This can include regulating emissions from vehicles and factories promoting energy sources and implementing urban planning measures to reduce pollution exposure [15].

Moreover looking at bad things in the air at the same time contributes to a better understanding of atmospheric processes and environmental dynamics. It lets researchers explore how bad things in the air move change and accumulate across regions and times. This comprehensive approach is crucial for tackling issues like urban air pollution, climate change and environmental sustainability [16]. Overall looking at bad things in the air at the same time offers a more realistic view of air quality compared to looking at one bad thing at a time. By considering interactions, combined effects and relationships among bad things in the air it improves the accuracy of air quality assessments enhances health risk evaluations and supports the development of effective solutions [17]. As air pollution continues to present challenges worldwide adopting this approach is essential, for advancing research improving prediction models and ensuring

better environmental and public health outcomes [18].

Deep learning is an advanced subset of machine learning that utilizes artificial neural networks with multiple hidden layers to model complex and high-dimensional data [19]. Unlike traditional machine learning approaches that rely heavily on manual feature extraction, deep learning algorithms automatically learn feature representations from raw data through hierarchical processing. These models are inspired by the biological structure of the human brain, where interconnected neurons process information and adapt based on experience. The increasing availability of large datasets, along with advancements in computational power such as Graphics Processing Units (GPUs), has significantly contributed to the rapid growth and success of deep learning in recent years. At the core of deep learning are artificial neural networks composed of input layers, hidden layers, and output layers. Each layer consists of nodes or neurons that are connected with weighted edges. During training, the model learns optimal weights through a process known as backpropagation, where errors are propagated backward and weights are updated using optimization algorithms such as gradient descent. Activation functions such as ReLU (Rectified Linear Unit), sigmoid, and tanh introduce non-linearity into the model, enabling it to learn complex relationships between inputs and outputs. The depth of the network, characterized by the number of hidden layers, allows deep learning models to extract increasingly abstract features at each level [20].

Several types of deep learning architectures have been developed to address different types of data and applications. Convolutional Neural Networks (CNNs) are widely used for image and spatial data processing due to their ability to automatically detect patterns such as edges, textures, and shapes through convolutional filters [21]. Recurrent Neural Networks (RNNs) are designed to handle sequential data by maintaining a memory of previous inputs, making them suitable for time-series analysis and natural language processing. However, traditional RNNs suffer from issues such as vanishing and exploding gradients, which limit their ability to learn long-term dependencies. To overcome these challenges, advanced architectures such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU) were introduced [22].

These models incorporate gating mechanisms that regulate the flow of information, allowing them to retain relevant information over longer sequences and improve prediction performance. Deep learning has found applications across various domains, including computer vision, speech recognition, natural language processing, healthcare, finance, and environmental monitoring. In the context of air quality prediction, deep learning models are particularly effective due to their ability to capture temporal dependencies and non-linear relationships in environmental data. Air pollution data often exhibit complex patterns influenced by multiple interacting factors such as emission sources, meteorological conditions, and seasonal variations. Models such as LSTM and GRU are well-suited for this task as they can learn from historical time-series data and predict future pollutant concentrations with high accuracy [23].

Furthermore, hybrid deep learning models, which combine two or more architectures, have

been proposed to enhance predictive performance [24]. For example, CNN-LSTM models can extract spatial features using convolutional layers and then model temporal dependencies using LSTM layers. Similarly, combinations such as BiLSTM-GRU or GRU-RNN can leverage the strengths of different architectures to improve generalization and robustness. These hybrid approaches are increasingly used in research to achieve better accuracy in complex prediction tasks such as air quality forecasting. Despite its advantages, deep learning also faces certain challenges. These models require large amounts of labeled data for effective training, which may not always be available. They are computationally expensive and require powerful hardware resources for training and deployment. Additionally, deep learning models are often considered as “black-box” systems, making it difficult to interpret their internal decision-making processes. Researchers are actively working on techniques such as explainable artificial intelligence (XAI) to address these limitations and improve model transparency [25].

Overall, deep learning represents a powerful and flexible approach for modeling complex systems and solving real-world problems. Its ability to automatically learn features, handle large datasets, and model non-linear relationships makes it highly suitable for applications such as air quality prediction. By integrating advanced architectures and hybrid models, deep learning continues to play a crucial role in the development of intelligent systems for environmental monitoring and decision-making [26].

CHAPTER 2- LITERATURE REVIEW

2 LITERATURE REVIEW

Air pollution has emerged as a major global environmental and public health concern, primarily driven by rapid industrialization, urbanization, and increasing vehicular emissions. Key atmospheric pollutants such as particulate matter ($\text{PM}_{2.5}$ and PM_{10}), carbon monoxide (CO), nitrogen dioxide (NO_2), and ground-level ozone (O_3) significantly affect human health and ecological systems [27]. Exposure to these pollutants is associated with severe respiratory and cardiovascular diseases, increased hospital admissions, and premature mortality. Consequently, accurate monitoring and prediction of air quality have become essential for effective environmental management and public health protection. Traditional air quality monitoring systems rely on fixed monitoring stations that collect real-time pollutant data [28]. However, these stations are often limited in number and spatial coverage, making it difficult to capture localized variations in air quality. This limitation has led to the growing need for intelligent predictive systems capable of estimating air quality in unmonitored regions using historical data and computational models.

Air quality assessment involves analyzing both pollutant concentrations and meteorological parameters, as their interactions significantly influence atmospheric behavior [29]. Particulate matter, particularly $\text{PM}_{2.5}$, consists of fine particles that can penetrate deep into the lungs and bloodstream, posing serious health risks [30]. PM_{10} , although larger in size, contributes to respiratory irritation and reduced lung function. Nitrogen dioxide (NO_2), primarily emitted from combustion processes, contributes to smog formation and respiratory illnesses. Carbon monoxide (CO) is a toxic gas that reduces oxygen transport in the bloodstream, while ozone (O_3), a secondary pollutant formed through photochemical reactions, is known to impair lung function and damage vegetation [31]. Earlier studies often focused on individual pollutants; however, recent research emphasizes the importance of multi-pollutant analysis. In real-world scenarios, humans are exposed to a mixture of pollutants that interact chemically and physically, leading to compounded health effects [32]. For example, nitrogen oxides react with volatile organic compounds under sunlight to produce ozone, while secondary particulate matter can form from gaseous precursors. Considering multiple pollutants simultaneously provides a more realistic and comprehensive understanding of air quality dynamics.

The advancement of machine learning and deep learning techniques has significantly improved the capability to model complex environmental systems. Unlike traditional models, which rely on predefined assumptions and linear relationships, deep learning algorithms can automatically learn intricate patterns and non-linear dependencies from large datasets [33]. Models such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU) have demonstrated superior performance in time-series prediction tasks, including air quality forecasting. Recurrent architectures, particularly LSTM and GRU, are well-suited for air quality prediction due to their ability to capture temporal dependencies in sequential data [34]. These models incorporate gating mechanisms that help retain relevant information over long time intervals, overcom-

ing issues such as vanishing gradients in traditional RNNs. Additionally, Bidirectional LSTM (Bi-LSTM) models enhance performance by processing data in both forward and backward directions, thereby capturing more contextual information [35].

Hybrid deep learning models have also gained attention for their improved predictive capabilities. For instance, CNN-LSTM models combine spatial feature extraction with temporal sequence learning, enabling better representation of complex environmental data [36]. Similarly, combinations such as GRU-LSTM and BiLSTM-GRU leverage the strengths of multiple architectures to enhance model robustness and accuracy [37]. Despite their advantages, deep learning models face certain challenges, including the requirement for large datasets, high computational costs, and limited interpretability. These models are often considered "black-box" systems, making it difficult to understand their internal decision-making processes. To address this limitation, recent studies have focused on explainable artificial intelligence (XAI) techniques to improve transparency and trust in predictive models [38].

Overall, the integration of multi-pollutant data with meteorological parameters, combined with advanced deep learning models, provides a powerful framework for accurate air quality prediction. This approach enhances the understanding of atmospheric processes, improves forecasting accuracy, and supports effective environmental policymaking and public health interventions. As air pollution continues to pose significant challenges globally, the adoption of intelligent predictive models remains essential for sustainable environmental management [39].

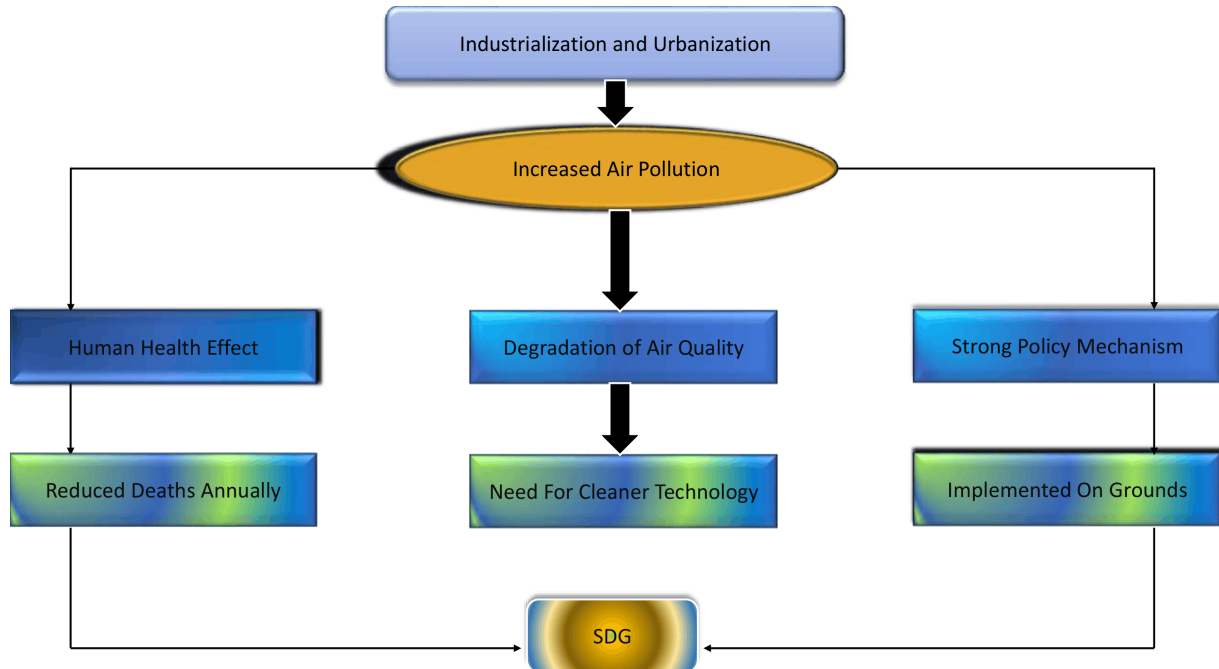


Figure 1: Impact of Urbanization on Air Quality and Sustainable Development

The flowchart presented in Figure 1 illustrated the relationship between urban development, air pollution, and sustainable outcomes. It begins with industrialization and urbanization, which act as the primary drivers of increased air pollution. The rise in pollution levels leads to the degra-

dation of air quality, affecting both environmental and human health conditions. As air quality deteriorates, it creates serious human health effects such as respiratory and cardiovascular diseases. This situation highlights the urgent need for cleaner and more sustainable technologies to control emissions and reduce pollution levels. On the policy side, strong regulatory mechanisms are required to address pollution effectively. However, the success of these policies depends on their proper implementation at the ground level. When both technological advancements and policy measures are effectively applied, they contribute to reducing mortality rates and improving overall environmental quality. Ultimately, these combined efforts align with the goals of sustainable development, ensuring a balance between urban growth and environmental protection.

CHAPTER 3- DATA

3 DATA

Chandni Chowk and R.K. Puram are two important places in Delhi where we measure air quality. Chandni Chowk is an old and crowded commercial area, with many people, cars and narrow streets. This makes the air quality very bad with levels of pollutants like particulate matter and nitrogen dioxide [40]. On the hand R.K. Puram is a well-planned residential area, with wider roads, less traffic and more green spaces. The air quality here is better than in Chandni Chowk. It is still affected by the overall pollution in Delhi. The weather also plays a role in determining air quality in these places. We collected data from air quality monitoring stations, which measure pollutants like particulate matter, nitrogen dioxide and ozone [41] [42]. This data helps us understand the patterns of air pollution in these areas and how they are affected by factors. We can use this data to develop models that predict air quality and help us make decisions about how to improve it. We are looking at two places Chandni Chowk and R.K. Puram, which are very different from each other. Chandni Chowk is a commercial area while R.K. Puram is a residential area with more green spaces. The data we collected includes information on pollutants like particulate matter, nitrogen dioxide and ozone. We also have data on weather conditions like wind speed and humidity [43]. We will use this data to understand the patterns of air pollution in these areas and to develop models that can predict air quality.

3.1 Study Area and Data Sources

The study focuses on two prominent air quality monitoring locations in Delhi, namely Chandni Chowk and R.K. Puram, which represent contrasting urban environments and pollution characteristics. Chandni Chowk is one of the oldest and most congested commercial areas in Delhi, characterized by dense population, narrow streets, and intense commercial and vehicular activity [44]. Due to continuous traffic movement, roadside emissions, and limited ventilation, this area experiences significantly higher levels of air pollution, particularly in terms of particulate matter ($PM_{2.5}$ and PM_{10}) and gaseous pollutants such as nitrogen dioxide (NO_2) and carbon monoxide (CO). The urban structure and heavy congestion make Chandni Chowk a critical hotspot for air quality analysis, especially for studying pollutant accumulation and peak pollution events [45]. In contrast, R.K. Puram is a well-planned residential locality with comparatively wider roads, better traffic regulation, and the presence of green spaces that contribute to improved air circulation [46].

Although it is still influenced by the overall pollution levels of Delhi, the pollutant concentrations in R.K. Puram are generally lower than those observed in Chandni Chowk. The area provides a comparatively cleaner environment, making it suitable for understanding moderate pollution levels and the impact of meteorological factors on pollutant dispersion. The comparison between these two locations enables a comprehensive analysis of how urban design, traffic density, and environmental conditions influence air quality [47]. The data used in this

study has been collected from official air quality monitoring stations that continuously record pollutant concentrations and meteorological parameters. The dataset includes major pollutants such as $PM_{2.5}$, PM_{10} , nitrogen dioxide (NO_2), carbon monoxide (CO), and ozone (O_3) [48]. These parameters are essential for understanding both the concentration levels of pollutants and the environmental factors that influence their behavior in the atmosphere. The data is typically obtained from government-authorized sources such as the Central Pollution Control Board (CPCB) and other environmental monitoring agencies, ensuring reliability and accuracy.

The collected dataset spans a large number of observations, enabling detailed statistical analysis and model development. Before analysis, the data undergoes preprocessing steps such as handling missing values, normalization, and feature selection to ensure data quality and consistency [49]. The inclusion of both pollutant and meteorological variables allows for a multi-dimensional analysis of air quality and supports the development of advanced predictive models using machine learning and deep learning techniques. Overall, the selected study areas and data sources provide a strong foundation for analyzing air pollution patterns and developing accurate forecasting models [50].

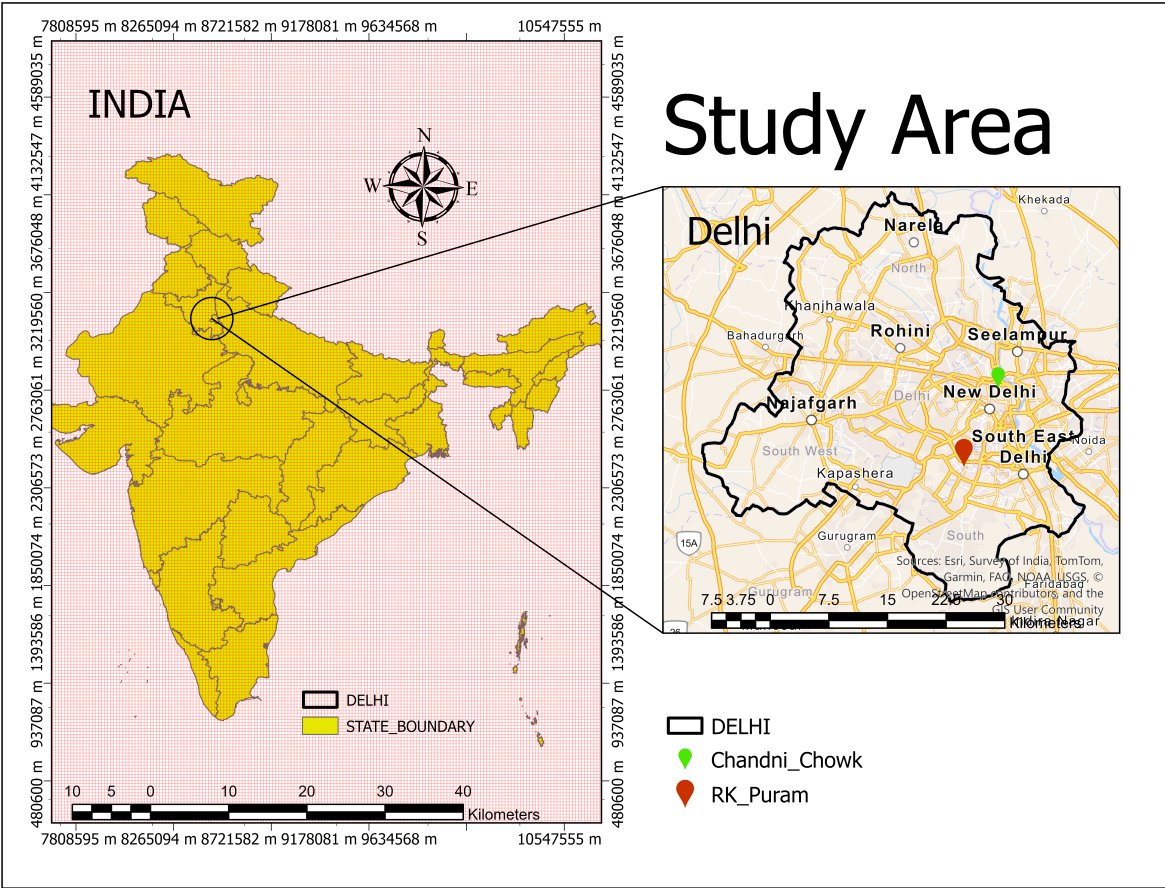


Figure 2: Study Area Map Showing Delhi and Selected Locations

3.2 Exploratory Data Analysis(EDA)

We are looking at two places in Delhi, Chandni Chowk and R.K. Puram Figure 2. These places are very different from each other with Chandni Chowk being a commercial area and R.K. Puram being a residential area. Chandni Chowk has people, cars and narrow streets which makes the air quality very bad [51]. R.K. Puram on the hand has wider roads, less traffic and more green spaces, which makes the air quality better [52]. We collected data on pollutants like particulate matter, nitrogen dioxide and ozone. We also collected data on weather conditions like wind speed and humidity. This data will help us understand the patterns of air pollution in these areas and how they are affected by factors. We can use this data to develop models that predict air quality and help us make decisions, about how to improve it.

3.2.1 Statistical Exploratory Analysis of Chandni Chowk

Table 1: Statistical Summary of Air Pollutants(Chandni Chowk)

Statistic	PM _{2.5}	PM ₁₀	NO ₂	CO	Ozone
Count	19143	19143	19143	19143	19143
Mean	106.78	212.39	54.94	1.71	26.93
Std Dev	65.77	135.13	37.56	1.03	24.65
Min	0.40	1.80	0.01	0.00	0.01
25%	71.04	116.77	33.44	1.06	11.42
Median (50%)	106.78	212.39	54.94	1.60	26.93
75%	106.78	243.40	58.27	1.86	28.34
Max	998.80	996.00	481.27	9.52	420.88

The statistical summary of air pollutants reveals significant variations in concentration levels across different parameters. PM_{2.5} and PM₁₀ exhibit high mean values of 106.78 and 212.39 respectively, indicating severe particulate pollution in the study area. The large standard deviation values (65.77 for PM_{2.5} and 135.13 for PM₁₀) suggest considerable fluctuations in pollutant concentrations over time Table 1. Nitrogen dioxide (NO₂) shows a moderate mean value of 54.94 with noticeable variability, reflecting its dependence on vehicular emissions and industrial activities. Carbon monoxide (CO) has a relatively low mean concentration of 1.71, but its maximum value of 9.52 indicates occasional spikes that may pose health risks. Ozone levels also demonstrate variability, with a mean of 26.93 and a maximum value reaching 420.88, highlighting the influence of photochemical reactions under varying environmental conditions. The presence of extremely high maximum values for PM_{2.5}, PM₁₀, and NO₂ indicates the occurrence of extreme pollution events, possibly due to factors such as seasonal changes, traffic congestion, or industrial emissions. Additionally, the difference between minimum and maximum values across all pollutants suggests a highly dynamic air quality pattern. Overall, the

data reflects a polluted environment with significant temporal variability, emphasizing the need for advanced predictive models and effective pollution control measures.

3.2.2 Statistical Exploratory Analysis of RK Puram

Table 2: Statistical summary of Air Pollutants(RK Puram)

Statistic	PM _{2.5}	PM ₁₀	NO ₂	CO	Ozone
Count	18642	18642	18642	18642	18642
Mean	118.25	211.34	40.00	1.39	39.25
Std	102.10	133.70	27.41	1.28	37.99
Min	1	1	0.1	0	0.1
25%	48	114.5	19.62	0.55	14.05
50%	86	190	33.7	0.97	25.38
75%	153	272.63	53.25	1.72	46.69
Max	975	998	220.95	9.59	431.45

The descriptive statistical analysis of the air quality dataset provides important insights into the distribution and variability of major pollutants. It can be observed that all pollutants have the same number of observations (18642), ensuring uniformity and consistency in the dataset Table 2. Among all pollutants, PM_{2.5} and PM₁₀ exhibit significantly higher mean concentrations of 118.25 and 211.34, respectively, indicating severe particulate pollution levels in the study area. The standard deviation values for PM_{2.5} (102.10) and PM₁₀ (133.70) are also quite large, reflecting high variability and frequent fluctuations in pollutant levels. The minimum values of pollutants are relatively low, whereas the maximum values reach extremely high levels, particularly for PM_{2.5} (975) and PM₁₀ (998), which indicates the occurrence of extreme pollution episodes. The interquartile ranges (25%, 50%, and 75%) further show that a large portion of the data is concentrated at lower values, but occasional spikes significantly increase the overall average. The median values for all pollutants are lower than their corresponding mean values, indicating a positively skewed distribution influenced by high outliers. For gaseous pollutants such as NO₂, CO, and Ozone, the mean values are comparatively moderate; however, their standard deviations still suggest noticeable variability. CO, in particular, shows a wide range from 0 to 9.59, highlighting intermittent high emission events. Similarly, Ozone exhibits substantial variation with a maximum value of 431.45, indicating strong photochemical activity under certain atmospheric conditions. Overall, the dataset demonstrates non-linear behavior, high dispersion, and the presence of extreme values across all pollutants. These characteristics indicate that traditional linear models may not be sufficient to capture the complex patterns present in the data. Therefore, advanced deep learning models are required to effectively model temporal dependencies and accurately forecast air pollutant concentrations.

3.2.3 Graphical Exploratory Analysis of Chandni Chowk

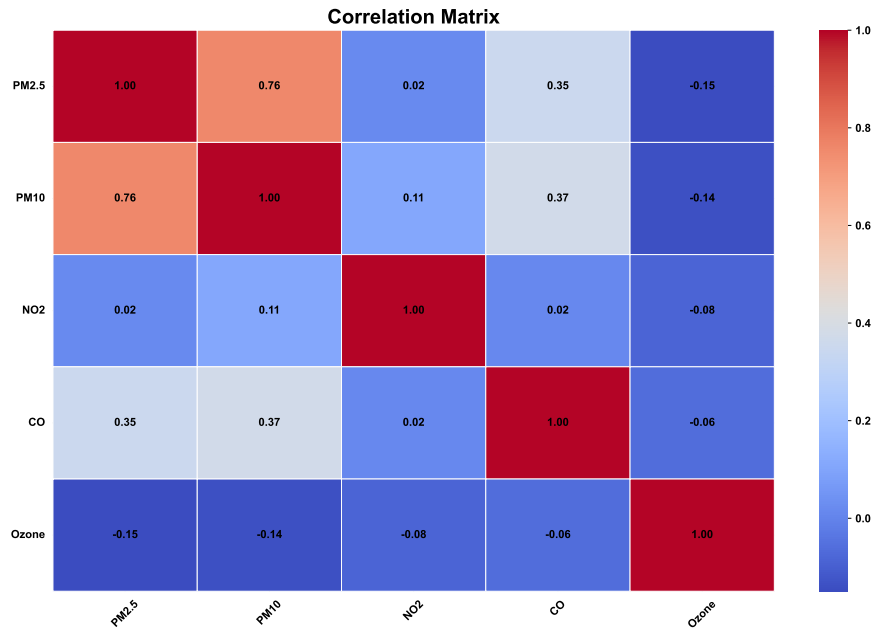


Figure 3: Distribution of air quality and meteorological variables of Chandni Chowk

PM_{2.5} & PM₁₀: Strong positive correlation (0.76). This suggests that both particulate matter types often rise and fall together, likely due to shared sources such as vehicular emissions and dust Figure 3 . PM_{2.5} & CO: Moderate positive correlation (0.35). This indicates that combustion-related sources (e.g., traffic and industrial activities) contribute to both pollutants. PM₁₀ & CO: Similar moderate correlation (0.37), reinforcing the hypothesis that these pollutants share common emission sources. NO₂: Very weak correlations with other pollutants (close to 0). This suggests that NO₂ levels are influenced by different factors, possibly localized traffic emissions. Ozone: Negative correlations with PM_{2.5} (-0.15), PM₁₀ (-0.14), NO₂ (-0.08), and CO (-0.06). This is typical since ozone formation is photochemical and often inversely related to primary pollutants.

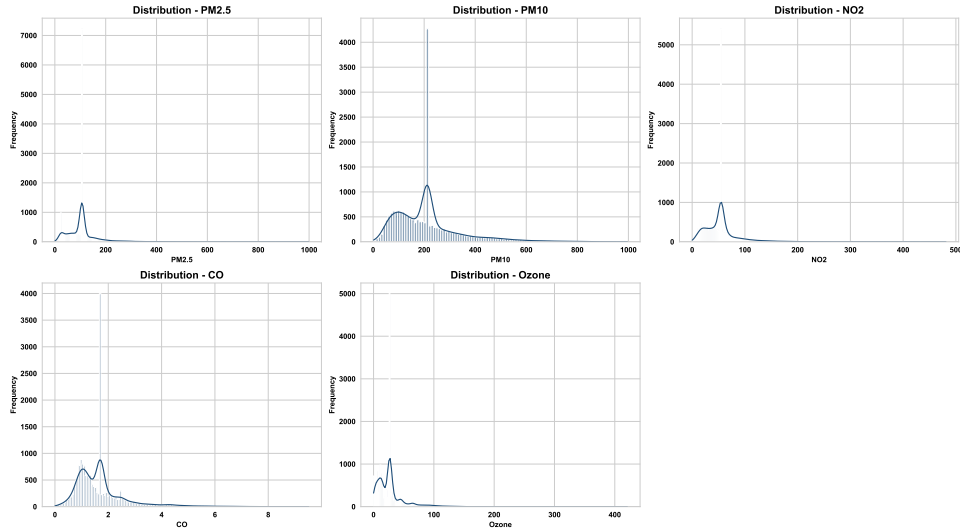


Figure 4: Distribution of air quality and meteorological variables of Chandni Chowk

PM_{2.5}: The distribution shows a wide spread with high frequency counts, suggesting frequent elevated concentrations. The data may be skewed toward higher values. PM₁₀: Displays a distribution pattern similar to PM_{2.5}, which is consistent with the strong correlation between the two pollutants. NO₂: The distribution appears more compact, with lower frequency counts, indicating fewer extreme values. CO: Shows a moderate spread with some peaks, suggesting episodic pollution events. Ozone: The distribution is relatively balanced but shows fewer extreme spikes compared to particulate matter pollutants Figure 4.

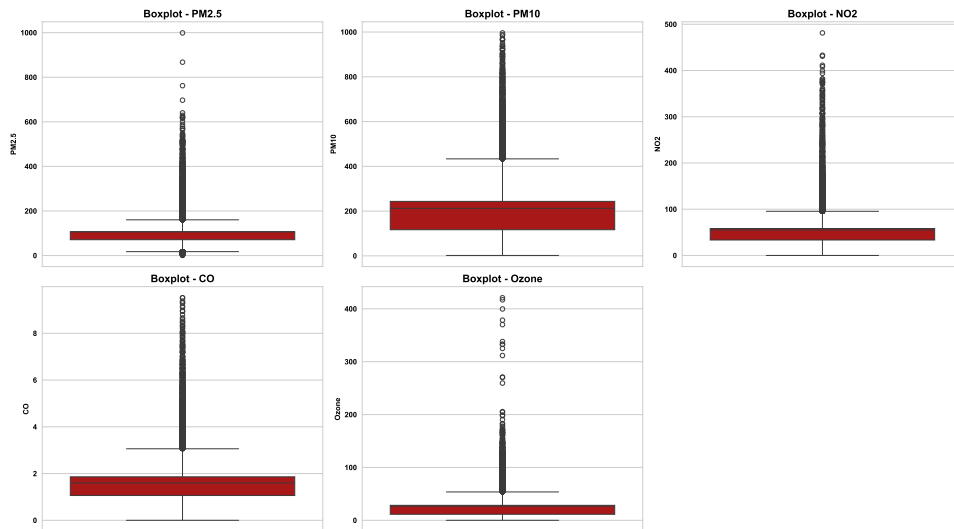


Figure 5: Distribution of air quality and meteorological variables of Chandni Chowk

PM_{2.5} & PM₁₀: Boxplots likely reveal significant variability and multiple outliers, consistent with their wide distributions. NO₂: Shows a narrower boxplot range with fewer outliers, indicating relatively stable concentration levels. CO: Displays a moderate spread with some outliers, reflecting episodic pollution events. Ozone: The boxplot suggests variability, but with

fewer extreme values compared to particulate matter pollutants Figure 5.

Overall Interpretation:Particulate Matter ($PM_{2.5}$, PM_{10}): Strongly correlated, highly variable, and dominant in overall pollution levels. CO: Moderately correlated with particulate matter; episodic spikes suggest combustion-related sources. NO_2 : Weakly correlated with other pollutants, relatively stable, likely associated with localized traffic emissions. Ozone: Inversely related to other pollutants, reflecting its nature as a secondary pollutant formed through photochemical reactions.

3.2.4 Graphical Exploratory Analysis of RK Puram

$PM_{2.5}$ & PM_{10} : Strong positive correlation (0.85). This indicates that both particulate matter pollutants tend to increase and decrease simultaneously, primarily due to shared emission sources such as vehicular exhaust, road dust, and industrial activities. $PM_{2.5}$ & CO: Moderately strong positive correlation (0.67). This suggests that combustion-related processes significantly contribute to both pollutants. PM_{10} & CO: Moderate positive correlation (0.63), further confirming common emission sources. NO_2 : Shows moderate correlations with $PM_{2.5}$ (0.39), PM_{10} (0.44), and CO (0.43), indicating partial dependence on traffic-related emissions but also influence from localized sources. Ozone: Negative correlations with $PM_{2.5}$ (-0.20), PM_{10} (-0.10), NO_2 (-0.27), and CO (-0.28). This behavior reflects the photochemical nature of ozone formation, which is often inversely related to primary pollutants Figure 6.

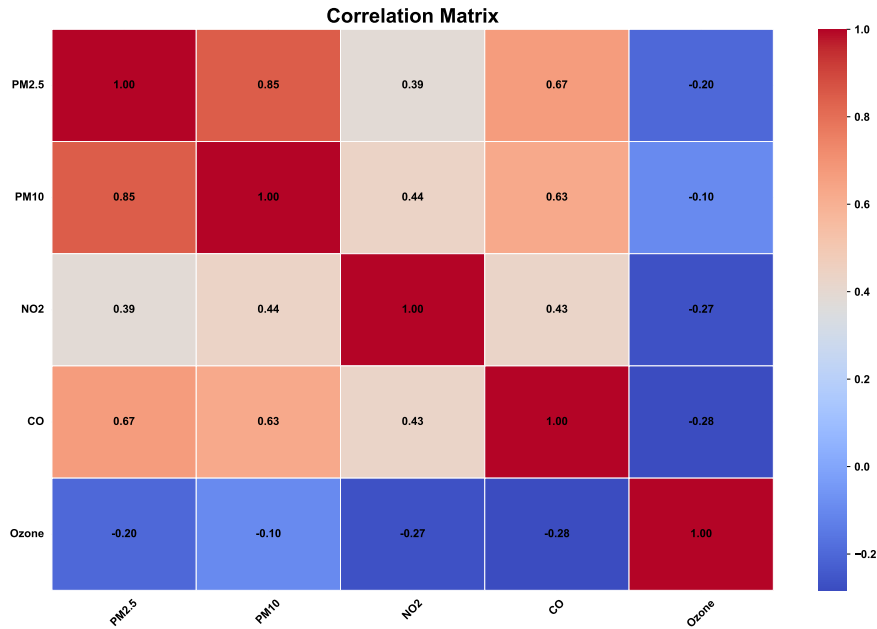


Figure 6: Distribution of air quality and meteorological variables of RK Puram

$PM_{2.5}$: The distribution is highly right-skewed, with most values concentrated at lower ranges and a few extreme high values, indicating occasional pollution spikes. PM_{10} : Exhibits a similar right-skewed distribution with a wider spread, suggesting higher variability and fre-

quent fluctuations in coarse particulate matter levels. NO_2 : Shows a moderately skewed distribution with most observations clustered in lower concentration ranges, indicating relatively stable levels with occasional increases. CO: Displays a slightly right-skewed distribution, with most values lying within a narrow range and a few higher values reflecting episodic emissions. Ozone: Presents a broader and more evenly spread distribution compared to other pollutants, indicating variability influenced by meteorological conditions such as sunlight and temperature Figure 7.

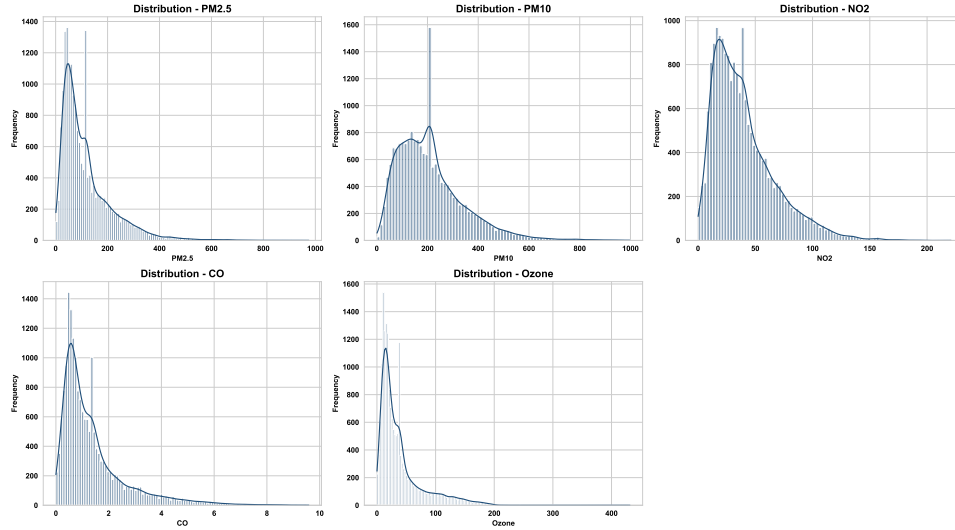


Figure 7: Distribution of air quality and meteorological variables of RK Puram

$\text{PM}_{2.5}$: The boxplot shows a wide interquartile range with numerous outliers on the higher end, indicating frequent extreme pollution events. PM_{10} : Displays a large spread and several high-value outliers, reflecting significant variability and occasional spikes in particulate matter concentration. NO_2 : Shows moderate variability with fewer outliers, suggesting relatively stable pollutant levels with limited extreme events. CO: The boxplot indicates a compact distribution with some outliers, highlighting generally consistent values with occasional increases. Ozone: Exhibits moderate spread with fewer extreme outliers, indicating variability but less frequent extreme spikes compared to particulate matter pollutants Figure 8.

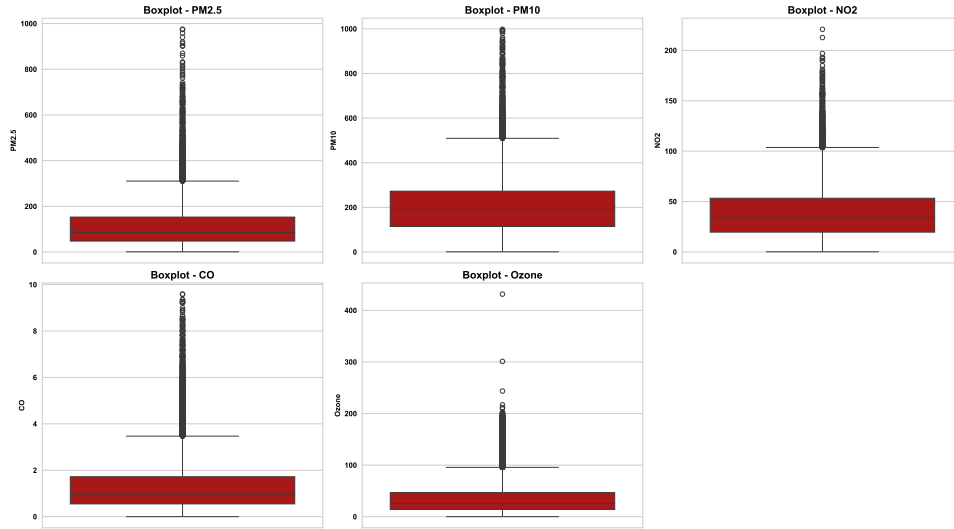


Figure 8: Distribution of air quality and meteorological variables of RK Puram

Overall Interpretation: Particulate Matter ($PM_{2.5}$, PM_{10}): Exhibits a strong positive correlation (0.85) and highly right-skewed distributions with significant outliers, indicating high variability and frequent extreme pollution events. These pollutants dominate overall air quality and primarily originate from common sources such as vehicular emissions, road dust, and industrial activities. CO: Shows a moderately strong positive correlation with $PM_{2.5}$ (0.67) and PM_{10} (0.63), confirming its association with combustion-based sources. The distribution is slightly right-skewed with occasional outliers, indicating episodic increases in emissions. NO_2 : Displays moderate correlations with $PM_{2.5}$ (0.39), PM_{10} (0.44), and CO (0.43), suggesting partial dependence on traffic emissions. Its distribution is relatively stable with fewer extreme outliers, indicating less variability compared to particulate matter. Ozone: Shows negative correlations with $PM_{2.5}$ (-0.20), NO_2 (-0.27), and CO (-0.28), reflecting its secondary photochemical formation. The distribution is more spread out, influenced by meteorological conditions such as solar radiation, with fewer extreme spikes.

CHAPTER 4- DEEP LEARNING MODELS

4 DEEP LEARNING MODELS

Deep learning models are machine learning methods. They use neural networks with several layers to learn complex patterns from large datasets. These models are effective in capturing relationships and time-based links in data. They are good for applications like predicting air quality. Common deep learning types include Recurrent Neural Networks and Long Short-Term Memory and Gated Recurrent Units and Bidirectional Long Short-Term Memory and Convolutional Neural Networks [4]. In this study deep learning models look at -pollutant data to improve prediction accuracy. Their ability to automatically pull out features and manage large-scale data is useful for monitoring. Deep learning models are really helpful because they can learn from datasets. They are good at finding patterns that're not easy to see. This makes them useful for predicting things like air quality [53].

4.1 Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNN) are deep learning models made to handle structured grid-like data especially spatial ones like images [54]. They use convolutional layers and pooling layers and fully connected layers to automatically pull out important features from the input. The convolution operation helps catch local patterns. Pooling reduces size and cuts down on computational complexity. CNNs are widely used in image classification and object detection and pattern recognition. In time-series applications, CNN extracts spatial and short-term temporal features. In air quality prediction, CNN identifies localized pollutant patterns and changes making it useful for feature extraction before feeding into sequential models. One key advantage of CNN is its ability to learn features by itself without manual work. The mathematical representation of convolution is given by:

$$y(i, j) = \sum_m \sum_n x(i - m, j - n) \cdot w(m, n) \quad (1)$$

where: $x(i, j)$ = input feature map $w(m, n)$ = convolution kernel (filter) $y(i, j)$ = output feature map m, n = kernel dimensions

4.2 Recurrent Neural Network (RNN)

Recurrent Neural Networks (RNN) are designed to handle sequential and time-dependent data by maintaining a hidden state that acts as memory [55]. Unlike traditional neural networks, RNNs allow information to persist across time steps, enabling them to capture temporal dependencies. They are widely used in sequence modeling tasks such as speech recognition, language translation, and time-series forecasting. However, standard RNNs suffer from the vanishing and exploding gradient problems, which make it difficult to learn long-term dependencies. Despite

this limitation, RNNs are useful for capturing short-term trends in sequential data and serve as a foundation for more advanced architectures like LSTM and GRU.

The mathematical equations are:

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b) \quad (2)$$

$$y_t = W_y h_t \quad (3)$$

where: h_t = hidden state at time t , h_{t-1} = previous hidden state, x_t = input at time t , W_h, W_x, W_y = weight matrices, b = bias term, and y_t = output at time t

4.3 Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) networks are a specialized type of RNN designed to overcome the limitations of traditional RNNs. LSTM uses memory cells along with three gating mechanisms: input gate, forget gate, and output gate [56]. These gates regulate the flow of information, allowing the network to retain important information and discard irrelevant data over long sequences. This makes LSTM highly effective in capturing long-term dependencies in time-series data. LSTM models are widely used in applications such as air quality forecasting, weather prediction, stock market analysis, and natural language processing. Their ability to handle long sequences and nonlinear relationships makes them one of the most powerful models for environmental data analysis.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (4)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (5)$$

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (6)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (7)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (8)$$

$$h_t = o_t \cdot \tanh(C_t) \quad (9)$$

where: f_t = forget gate, i_t = input gate, o_t = output gate, C_t = cell state, $\tilde{C}_t$ = candidate cell state, h_t = hidden state, x_t = input at time t , W_f, W_i, W_o, W_c = weight matrices, b_f, b_i, b_o, b_c = bias terms, and σ = sigmoid activation function

4.4 Gated Recurrent Unit (GRU)

Gated Recurrent Unit (GRU) is an improved variant of RNN that simplifies the architecture of LSTM while maintaining comparable performance. GRU uses two main gates: update gate and reset gate, which control the flow of information [57]. This simplified structure reduces the number of parameters, making GRU computationally efficient and faster to train compared to LSTM. GRU is particularly suitable for real-time applications and large datasets. It is widely used in time-series prediction, speech processing, and air quality modeling. Despite having a simpler structure, GRU performs effectively in capturing both short-term and long-term dependencies in sequential data.

$$z_t = \sigma(W_z[h_{t-1}, x_t]) \quad (10)$$

$$r_t = \sigma(W_r[h_{t-1}, x_t]) \quad (11)$$

$$\tilde{h}_t = \tanh(W[r_t \cdot h_{t-1}, x_t]) \quad (12)$$

$$h_t = (1 - z_t) \cdot h_{t-1} + z_t \cdot \tilde{h}_t \quad (13)$$

where: z_t = update gate , r_t = reset gate , $\tilde{h}_t$ = candidate hidden state , h_t = current hidden state , h_{t-1} = previous hidden state , x_t = input , W_z, W_r, W = weight matrices , and σ = sigmoid function

4.5 Bidirectional Long Short-Term Memory (Bi-LSTM)

Bidirectional Long Short-Term Memory (Bi-LSTM) is an extension of the LSTM model that processes input sequences in both forward and backward directions. This dual processing allows the model to capture information from both past and future contexts, improving prediction accuracy. Bi-LSTM consists of two LSTM layers: one processes the sequence from start to end, while the other processes it in reverse order [58]. This makes it highly effective for tasks where full sequence context is important. Bi-LSTM is widely used in applications such as speech recognition, sentiment analysis, and environmental forecasting. In air quality prediction, Bi-LSTM enhances model performance by utilizing complete temporal information, leading to more accurate and reliable predictions.

$$z_t = \sigma(W_z[h_{t-1}, x_t]) \quad (14)$$

$$r_t = \sigma(W_r[h_{t-1}, x_t]) \quad (15)$$

$$\tilde{h}_t = \tanh(W[r_t \cdot h_{t-1}, x_t]) \quad (16)$$

$$h_t = (1 - z_t) \cdot h_{t-1} + z_t \cdot \tilde{h}_t \quad (17)$$

where: z_t = update gate , r_t = reset gate , $\tilde{h}_t$ = candidate hidden state , h_t = current hidden

state , h_{t-1} = previous hidden state , x_t = input , W_z, W_r, W = weight matrices , and σ = sigmoid function

Training and Optimization

Deep learning models are trained using optimization techniques such as gradient descent and backpropagation. The objective is to minimize the loss function, which measures the difference between predicted and actual values.

A commonly used loss function for regression problems is Mean Squared Error (MSE): [59]

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (18)$$

where y_i is the actual value and $\hat{y}_i$ is the predicted value. Optimization algorithms such as Adam, RMSProp, and Stochastic Gradient Descent (SGD) are used to update model parameters and improve performance [60]. Overall, deep learning models provide powerful tools for handling complex, high-dimensional data and have become essential in modern artificial intelligence applications.

The flowchart presented in Figure 9. illustrates the overall methodology adopted for air quality prediction using deep learning techniques. The process is systematically divided into three major stages: training, validation, and testing. Initially, the collected dataset undergoes pre-processing and feature extraction to ensure data quality and consistency. In the training phase, the deep learning model is developed and trained using historical pollutant and meteorological data. The validation stage is used to fine-tune model parameters and prevent overfitting by evaluating performance on unseen data during training. Finally, in the testing phase, the model is assessed using completely new data to measure its generalization capability and prediction accuracy. This structured workflow ensures robust model development and reliable performance evaluation for air quality forecasting.

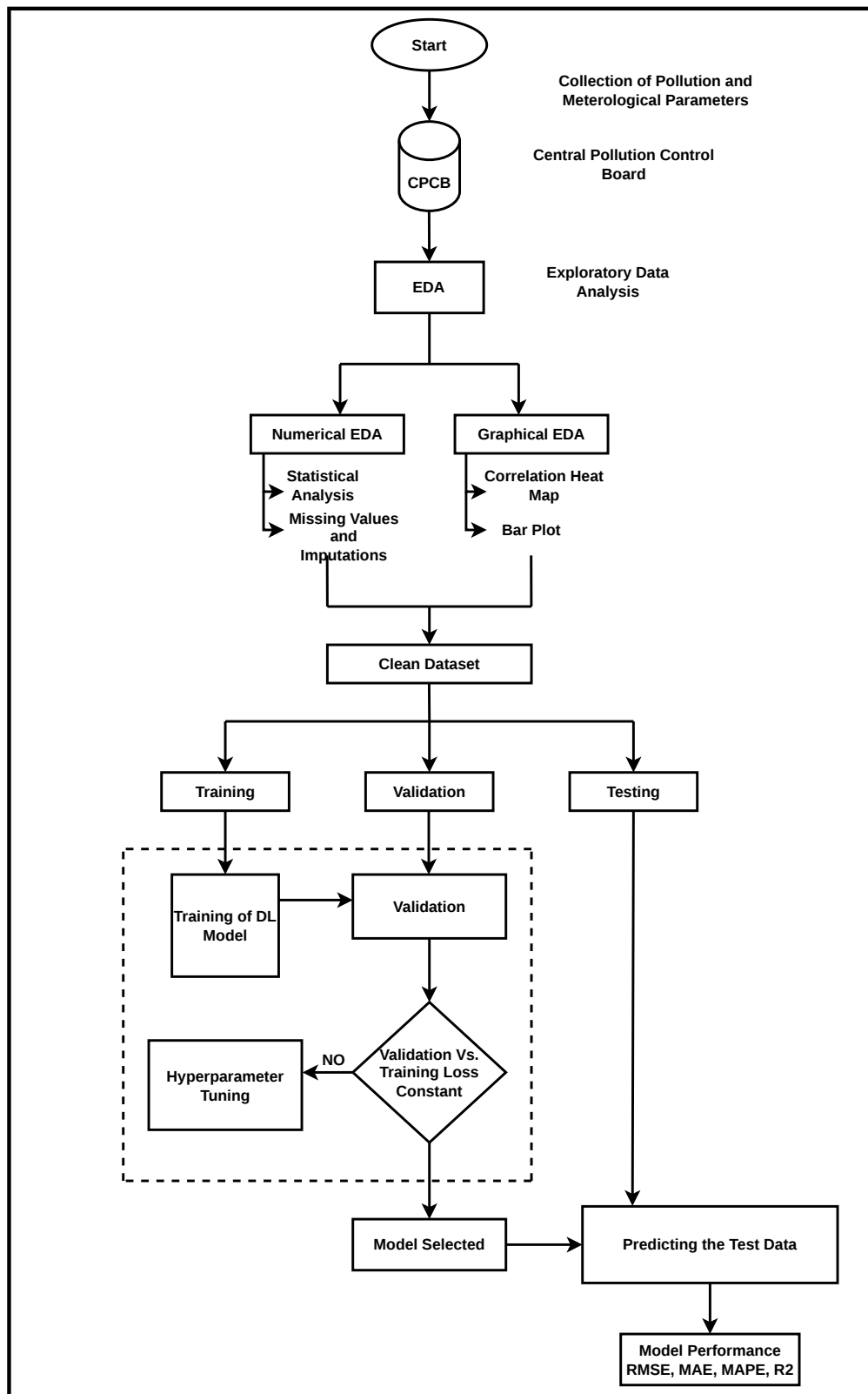


Figure 9: Deep Learning Models Workflow for Air Quality Prediction with Training, Validation, and Testing Data

CHAPTER 5- METHODOLOGY

5 METHODOLOGY

The flowchart illustrated in Figure 10 presents a comprehensive framework for air quality prediction using deep learning techniques. The process begins with data collection, which includes major air pollutants such as PM2.5, PM10, NO₂, CO, and Ozone, along with relevant meteorological parameters. In the preprocessing stage, data cleaning is performed to remove noise and inconsistencies, followed by missing data imputation to handle incomplete records. Subsequently, data transformation techniques are applied to normalize and scale the dataset for improved model performance.

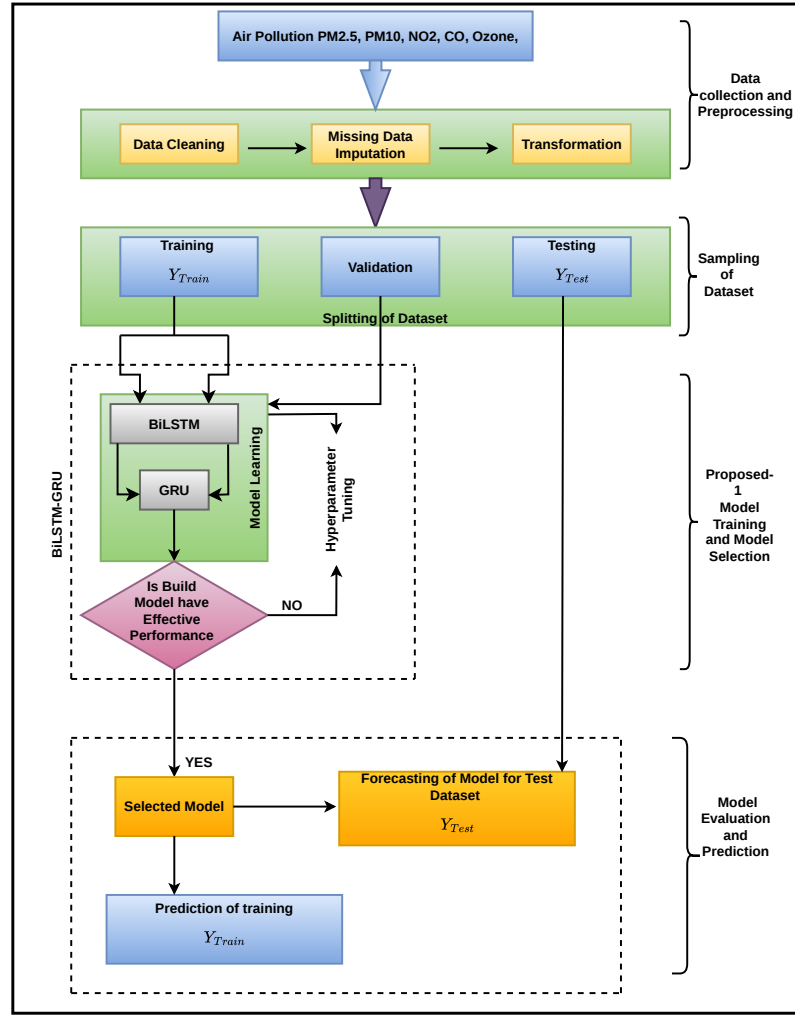


Figure 10: Proposed Framework for Air Quality Prediction using Hybrid BiLSTM-GRU Deep Learning Models

After preprocessing, the dataset is divided into three subsets: training (Y_{Train}), validation, and testing (Y_{Test}). The training dataset is used to build deep learning models, including Bidirectional Long Short-Term Memory (Bi-LSTM) networks and Gated Recurrent Unit (GRU). These models are trained to capture temporal and spatial dependencies in air quality data. The

validation dataset plays a crucial role in hyperparameter tuning and model optimization, ensuring that the model does not overfit.

An iterative model learning process is adopted, where the model performance is continuously evaluated. If the model does not achieve effective performance, hyperparameter tuning is performed, and the training process is repeated. Once the model satisfies the performance criteria, it is selected as the final model. The selected model is then used to generate predictions for both training (Y_{Train}) and testing (Y_{Test}) datasets. Finally, the forecasting results obtained from the testing dataset are used to evaluate the overall accuracy and generalization capability of the proposed model.

Let the multivariate time series dataset be defined as:

$$X_t = [x_t^{(1)}, x_t^{(2)}, x_t^{(3)}, x_t^{(4)}, x_t^{(5)}] \quad (19)$$

where: PM2.5 concentration, PM10 concentration, NO₂ concentration, CO concentration, O₃ concentration. Given a sequence of past observations of length T , the objective is to predict future pollution levels:

$$Y_{t+1} = f(X_{t-T+1}, X_{t-T+2}, \dots, X_t) \quad (20)$$

Each feature is normalized using Min-Max scaling:

$$\tilde{x}^{(i)} = \frac{x_t^{(i)} - x_{\min}^{(i)}}{x_{\max}^{(i)} - x_{\min}^{(i)}} \quad (21)$$

The Bidirectional LSTM processes the sequence in both forward and backward directions.

$$\vec{h}_t = \text{LSTM}_f(x_t, \overrightarrow{ht-1}) \quad (22)$$

$$\overleftarrow{h}_t = \text{LSTM}_b(x_t, \overleftarrow{ht+1}) \quad (23)$$

$$h_t^{Bi} = [\vec{h}_t; \overleftarrow{h}_t] \quad (24)$$

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (25)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (26)$$

$$\tilde{C}_t = \tanh(W_c[ht-1, x_t] + b_c) \quad (27)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (28)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (29)$$

$$h_t = o_t \odot \tanh(C_t) \quad (30)$$

The output from BiLSTM is passed to the GRU layer:

$$z_t = \sigma(W_z h_t^{Bi} + U_z h_{t-1}^{GRU} + b_z) \quad (31)$$

$$r_t = \sigma(W_r h_t^{Bi} + U_r h_{t-1}^{GRU} + b_r) \quad (32)$$

$$\tilde{h}t = \tanh(W_h h_t^{Bi} + U_h(r_t \odot h_{t-1}^{GRU} + b_h)) \quad (33)$$

$$h_t^{GRU} = (1 - z_t) \odot h_{t-1}^{GRU} + z_t \odot \tilde{h}t \quad (34)$$

The final prediction is obtained using a fully connected layer:

$$\hat{Y}_{t+1} = W_d h_t^{GRU} + b_d \quad (35)$$

Mean Squared Error (MSE) is used for training:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2 \quad (36)$$

Root Mean Square Error (RMSE):

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2} \quad (37)$$

Mean Absolute Error (MAE):

$$MAE = \frac{1}{N} \sum_{i=1}^N |Y_i - \hat{Y}_i| \quad (38)$$

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum (Y_i - \hat{Y}_i)^2}{\sum (Y_i - \bar{Y})^2} \quad (39)$$

Algorithm 1 Hybrid BiLSTM-GRU for Air Pollution Prediction

0: **Input:** Multivariate time series data $X = \{PM2.5, PM10, NO_2, CO, O_3\}$, sequence length T

0: **Output:** Predicted pollution level $\hat{Y}_{t+1}$

0: **Data Preprocessing:**

for each feature $x^{(i)}$ in X **do**

 Normalize using Min-Max scaling:

$$\tilde{x}_t^{(i)} = \frac{x_t^{(i)} - x_{\min}^{(i)}}{x_{\max}^{(i)} - x_{\min}^{(i)}}$$

end for

Construct input sequences:

$$X_t = \{\tilde{x}_{t-T+1}, \tilde{x}_{t-T+2}, \dots, \tilde{x}_t\}$$

BiLSTM Layer:

for $t = 1$ to T **do**

 Forward pass: $\vec{h}_t = \text{LSTM}_f(x_t, \vec{h}_{t-1})$

 Backward pass: $\overleftarrow{h}_t = \text{LSTM}_b(x_t, \overleftarrow{h}_{t+1})$

 Concatenate: $h_t^{Bi} = [\vec{h}_t; \overleftarrow{h}_t]$

end for

GRU Layer:

for $t = 1$ to T **do**

 Update gate: $z_t = \sigma(W_z h_t^{Bi} + U_z h_{t-1} + b_z)$

 Reset gate: $r_t = \sigma(W_r h_t^{Bi} + U_r h_{t-1} + b_r)$

 Candidate activation: $\tilde{h}_t = \tanh(W_h h_t^{Bi} + U_h(r_t \odot h_{t-1}) + b_h)$

 Hidden state update: $h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$

end for

Dense Output Layer:

$$\hat{Y}_{t+1} = W_d h_T + b_d$$

Loss Computation:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2$$

Model Training:

Update weights using backpropagation and gradient descent

Return: $\hat{Y}_{t+1}$

CHAPTER 6- EXPERIMENTAL SETUP

6 EXPERIMENTAL SETUP

The experimental setup for hourly multi-pollutant air quality forecasting using an LSTM-based deep learning model consists of a computational environment designed to efficiently process large-scale time-series data and train deep learning architectures. The setup includes both hardware and software components that ensure smooth model training, evaluation, and visualization.

6.1 Hardware and Software Required

The hardware configuration provides sufficient computational capability to handle large-scale time-series datasets and iterative model training processes Table 3. The use of a multi-core CPU and adequate RAM ensures efficient preprocessing and feature engineering, while optional GPU acceleration significantly reduces training time for deep learning models such as LSTM. On the software side, Python is used as the primary programming language due to its extensive ecosystem for data analysis and machine learning. Libraries such as NumPy and Pandas facilitate efficient data manipulation, while Matplotlib and Seaborn are used for visualization of temporal trends and model performance. The LSTM model is implemented using TensorFlow/Keras or PyTorch, which provide robust tools for designing and training deep neural networks. Additionally, Scikit-learn is utilized for data preprocessing, scaling, and evaluation metrics. Overall, the integrated hardware and software setup ensures reliable, scalable, and reproducible experimentation for hourly multi-pollutant air quality forecasting.

Table 3: Hardware and Software Configuration for Experimental Setup

Component	Specification
Hardware Configuration	
Processor (CPU)	Intel Core i5/i7 or equivalent
RAM	8 GB / 16 GB
Storage	256 GB / 512 GB SSD
GPU (Optional)	NVIDIA GPU (e.g., GTX 1650 / RTX series)
Operating System	Windows 10 / Linux (Ubuntu)
Software Configuration	
Programming Language	Python (3.8 or above)
Libraries	NumPy, Pandas, Matplotlib, Seaborn
Deep Learning Framework	TensorFlow / Keras / PyTorch
IDE/Environment	Jupyter Notebook / Google Colab / VS Code
Data Processing Tools	Scikit-learn
Visualization Tools	Matplotlib, Seaborn

6.2 Parameters Settings of Deep Learning Models

The performance metrics used in this study provide a comprehensive quantitative evaluation of prediction accuracy. RMSE and MAE measure the average magnitude of prediction errors, with RMSE giving higher weight to larger errors. MAPE expresses the prediction error in percentage form, making it easier to interpret relative error. The coefficient of determination (R^2) evaluates how well the predicted values explain the variance in the observed data, with values closer to 1 indicating better model performance Table 4.

Table 4: Performance Metrics for Quantitative Evaluation of Prediction Accuracy

Metric	Formula	Description of Variables
RMSE	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$	y_i = Actual value, $\hat{y}_i$ = Predicted value, n = Number of observations
MAE	$MAE = \frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $	y_i = Actual value, $\hat{y}_i$ = Predicted value, n = Total number of data points
MAPE	$MAPE = \frac{100}{n} \sum_{i=1}^n \left \frac{y_i - \hat{y}_i}{y_i} \right $	y_i = Actual value, $\hat{y}_i$ = Predicted value, n = Number of observations
R^2	$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$	y_i = Actual value, $\hat{y}_i$ = Predicted value, $\bar{y}$ = Mean of actual values, n = Number of samples

CHAPTER 7- RESULTS

7 RESULTS

7.1 Results and Analysis

Traditional Deep Learning Models Comparison of Chandni Chowk and RK Puram

7.1.1 Overall Model Performance Comparison Table of Chandni Chowk

The comparative evaluation of traditional deep learning models, namely RNN, BiLSTM, LSTM, GRU, and CNN, was conducted using multiple performance metrics including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and R-squared (R^2). RNN achieved the best overall performance, securing the top rank across most evaluation metrics Table 5. It recorded the lowest MSE (2924.48), RMSE (39.21), MAE (21.88), and MAPE (146.75), along with the highest R^2 value (0.6748). This indicates that RNN provides the most accurate and reliable predictions among all models. BiLSTM demonstrated moderate performance. Although it effectively captures bidirectional dependencies, it showed slightly higher error values compared to RNN, with an MSE of 3176.91 and RMSE of 40.77. Its R^2 value (0.6168) suggests reasonable predictive capability but lower accuracy than RNN. LSTM performed similarly to BiLSTM, with slightly better MAPE (147.36), indicating improved percentage-based prediction accuracy. However, its error metrics (MSE: 3277.46, RMSE: 41.36) were higher than RNN, placing it in the mid-performance range. GRU showed competitive performance with an R^2 value of 0.6504, which is relatively close to RNN. However, its higher MSE (3344.30) and RMSE (41.57) indicate slightly reduced prediction accuracy compared to RNN. CNN performed the worst among all models, exhibiting the highest error values (MSE: 3953.38, RMSE: 46.74, MAE: 28.87, MAPE: 181.57) and the lowest R^2 value (0.4534).

Table 5: Comparison of Traditional Deep Learning Models based on Evaluation Metrics

Rank	Model	Avg_MSE	Avg_RMSE	Avg_MAE	Avg_MAPE	Avg_R2
1	RNN	2924.48	39.21	21.88	146.75	0.6748
2	BiLSTM	3176.91	40.77	22.26	159.56	0.6168
3	LSTM	3277.46	41.36	22.80	147.36	0.6322
4	GRU	3344.30	41.57	22.93	159.14	0.6504
5	CNN	3953.38	46.74	28.87	181.57	0.4534

This suggests that CNN is less suitable for time-series air pollution prediction compared to recurrent-based models. Overall, Recurrent models (RNN, LSTM, GRU, BiLSTM) outperform convolutional models (CNN) in time-series prediction tasks. Among them, RNN emerges as the most effective model for this dataset, offering the highest accuracy and lowest prediction error.

7.1.2 Overall Models Performance Comparison Table of RK Puram

The performance of deep learning models was evaluated using key metrics including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and R-squared (R^2) Table 6. BiLSTM achieved the best overall performance, securing the first rank. It recorded the lowest MSE (1105.34) and RMSE (25.68), along with a high R^2 value of 0.8632. This indicates that BiLSTM effectively captures both past and future dependencies in time-series data, resulting in highly accurate predictions. LSTM performed closely behind BiLSTM with comparable error values (MSE: 1110.32, RMSE: 25.79) and the lowest MAE (15.77). Its R^2 value (0.8614) reflects strong predictive capability, making it a reliable model for sequential data analysis. RNN showed stable performance with moderate error values (MSE: 1140.45, RMSE: 26.32). Although it performs well, its slightly lower R^2 value (0.8533) indicates that it is less effective than LSTM-based models in capturing long-term dependencies. GRU demonstrated reasonable performance with an R^2 value of 0.8459. However, its higher error metrics (MSE: 1224.57, RMSE: 27.14) suggest that it is slightly less accurate compared to LSTM and BiLSTM models. CNN again performed the worst among all models, with significantly higher error values (MSE: 2266.94, RMSE: 37.17, MAE: 24.42) and the lowest R^2 value (0.7004).

Table 6: Comparison of Traditional Deep Learning Models based on Evaluation Metrics

Rank	Model	Avg_MSE	Avg_RMSE	Avg_MAE	Avg_MAPE	Avg_R2
1	BiLSTM	1105.34	25.68	15.99	1370177.63	0.8632
2	LSTM	1110.32	25.79	15.77	1424858.32	0.8614
3	RNN	1140.45	26.32	15.97	1564891.33	0.8533
4	GRU	1224.57	27.14	16.50	1267977.47	0.8459
5	CNN	2266.94	37.17	24.42	1808447.76	0.7004

This indicates that CNN is not well-suited for time-series prediction tasks in this context. Overall, Bidirectional and recurrent architectures outperform convolutional models for air pollution prediction. Among all models, BiLSTM provides the most accurate and reliable results, making it the preferred choice for this dataset.

7.1.3 Featurewise Model Comparison Table of Chandni Chowk

The performance of deep learning models including LSTM, BiLSTM, GRU, RNN, and CNN for air pollutant prediction at the Chandni Chowk location is presented in Table 7. For $PM_{2.5}$, the RNN model achieved the best performance with the lowest RMSE (53.35) and highest R^2 value (0.755), indicating its effectiveness in capturing temporal variations in highly polluted urban environments. BiLSTM also showed competitive performance, while CNN exhibited the weakest results. In the case of PM_{10} , RNN again outperformed other models with the highest R^2 value of 0.672. LSTM and BiLSTM showed similar performance trends, whereas CNN resulted

in significantly higher error values. For NO_2 , GRU performed slightly better than other models with the highest R^2 value (0.552), followed closely by RNN and LSTM. CNN once again showed poor predictive capability. For CO prediction, RNN achieved the best results with the lowest RMSE (0.686) and highest R^2 value (0.737), indicating strong performance in modeling gaseous pollutants in dense traffic conditions. In the case of Ozone, GRU demonstrated superior performance with the highest R^2 value (0.682), while CNN remained the least effective model.

Overall, it is observed that RNN and GRU perform better for the Chandni Chowk dataset, likely due to their ability to handle highly fluctuating and noisy data typical of congested urban regions. CNN consistently shows the weakest performance.

7.1.4 Featurewise Model Comparison Table of RK Puram

Table 8 presents the performance comparison of models for the RK Puram location. For $\text{PM}_{2.5}$, BiLSTM achieved the best performance with the lowest RMSE (42.63) and highest R^2 value (0.883), indicating improved prediction accuracy in a relatively stable and planned residential environment. In the case of PM_{10} , RNN slightly outperformed other models with the highest R^2 value of 0.852, followed closely by LSTM and BiLSTM. CNN again exhibited the poorest performance. For NO_2 , BiLSTM provided the best results with the highest R^2 value (0.852) and lowest RMSE (11.52), demonstrating its effectiveness in learning bidirectional temporal dependencies. For CO prediction, BiLSTM achieved the highest R^2 value (0.879), followed by LSTM and RNN. However, the extremely high MAPE values indicate possible data scaling issues or near-zero values, making MAPE less reliable for this pollutant. For Ozone, BiLSTM again demonstrated superior performance with the highest R^2 value (0.851), followed by LSTM and GRU. Overall, BiLSTM performs best for the RK Puram dataset, likely due to the relatively smoother and less noisy data patterns compared to Chandni Chowk. All models show improved performance, while CNN continues to underperform.

7.1.5 Traditional Model Comparison of Chandni Chowk

The MAE results show that the RNN model has the lowest error, indicating the highest accuracy among all models. GRU and LSTM also perform well with slightly higher error values. BiLSTM shows moderate performance, while CNN has the highest MAE, indicating poor prediction accuracy. Lower MAE values suggest that RNN and GRU are more reliable for this dataset Figure 11.

Table 7: Performance Comparison of Deep Learning Models Across Environmental Parameters of Chandni Chowk

Pollutant	LSTM			BiLSTM			GRU			RNN			CNN		
	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2	RMSE	MAE	R^2
PM2.5	57.98	29.15	21.61	0.710	55.58	27.36	20.57	0.734	58.67	29.04	21.86	0.703	59.85	33.61	0.691
PM10	111.09	60.67	20.07	0.635	109.97	59.61	21.47	0.643	112.37	61.95	20.60	0.627	122.01	77.33	0.560
NO2	19.00	12.09	157.03	0.536	19.43	12.10	149.08	0.514	18.66	11.50	153.61	0.552	24.44	15.93	0.232
CO	0.829	0.471	73.35	0.616	0.915	0.527	66.75	0.532	0.747	0.406	80.39	0.688	0.929	0.549	0.518
Ozone	17.93	11.63	464.72	0.664	18.00	11.73	539.93	0.661	17.45	11.77	519.26	0.682	26.49	16.95	0.266

Table 8: Performance Comparison of Deep Learning Models Across Environmental Parameters of RK Puram

Pollutant	LSTM				BiLSTM				GRU				RNN				CNN			
	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2
PM2.5	43.06	25.05	20.22	0.880	42.63	25.15	20.06	0.883	46.24	26.88	19.59	0.862	44.58	24.88	19.35	0.872	62.29	38.97	30.30	0.750
PM10	57.41	36.21	19.60	0.851	57.58	37.39	20.37	0.850	59.55	37.28	18.77	0.840	57.19	36.53	20.25	0.852	80.96	54.61	25.60	0.704
NO2	11.58	7.76	18.12	0.850	11.52	7.75	17.71	0.852	12.52	8.11	16.61	0.825	11.79	7.80	18.36	0.845	17.15	12.21	28.98	0.672
CO	0.558	0.358	7124132.07	0.877	0.553	0.363	6850738.84	0.879	0.590	0.372	6339720.70	0.863	0.572	0.362	7824286.53	0.871	0.842	0.572	9042001.92	0.721
Ozone	16.35	9.47	101.60	0.848	16.16	9.28	91.16	0.851	16.80	9.87	111.68	0.839	17.47	10.29	112.16	0.826	24.60	15.71	152.02	0.656

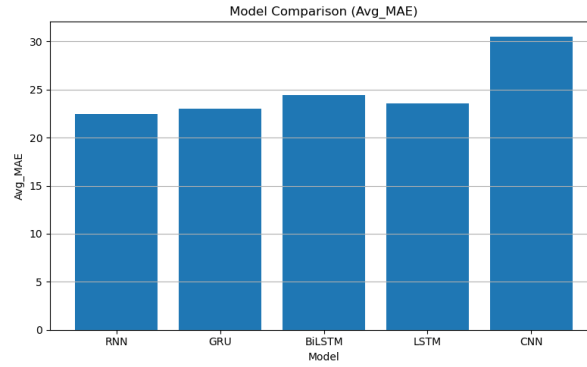


Figure 11: Model Comparison based on Average MAE

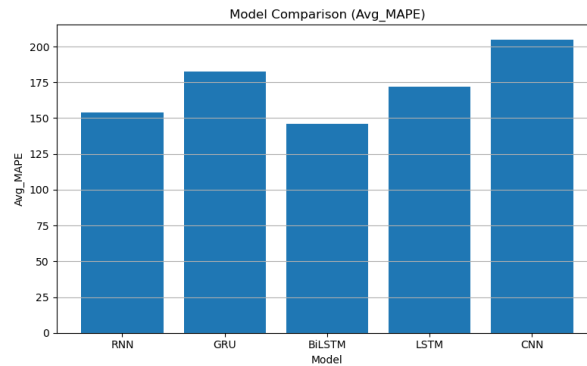


Figure 12: Model Comparison based on Average MAPE

The MAPE comparison indicates that BiLSTM achieves the lowest percentage error, making it the best model in terms of relative accuracy. RNN also performs well, followed by LSTM and GRU with moderate values. CNN again shows the highest MAPE, reflecting poor prediction performance. This suggests that BiLSTM is effective in minimizing percentage-based prediction errors Figure 12.

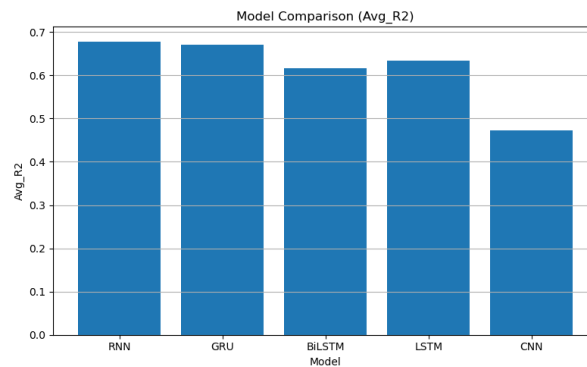


Figure 13: Model Comparison based on Average R^2 Score

The R^2 score comparison indicates that the RNN model achieves the highest value, followed closely by GRU and LSTM. This suggests that these models are better at explaining the variance in the dataset and provide more accurate predictions. BiLSTM shows moderate performance, while CNN has the lowest R^2 value, indicating poor model fitting and lower prediction reliability Figure 13.

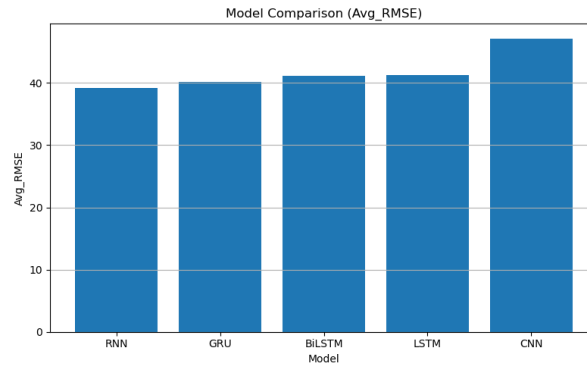


Figure 14: Model Comparison based on Average RMSE

The RMSE comparison shows that the RNN model has the lowest error, indicating better prediction accuracy. GRU, LSTM, and BiLSTM exhibit slightly higher RMSE values, reflecting moderate performance. CNN again shows the highest RMSE, indicating the largest prediction error. Lower RMSE values confirm that RNN and GRU are more effective models for this dataset. Based on all evaluation metrics, RNN and GRU consistently perform better across different error measures, while CNN shows the weakest performance. BiLSTM performs well in percentage-based metrics, making it suitable for specific forecasting scenarios Figure 14.

7.1.6 Traditional Model Comparison of RK Puram

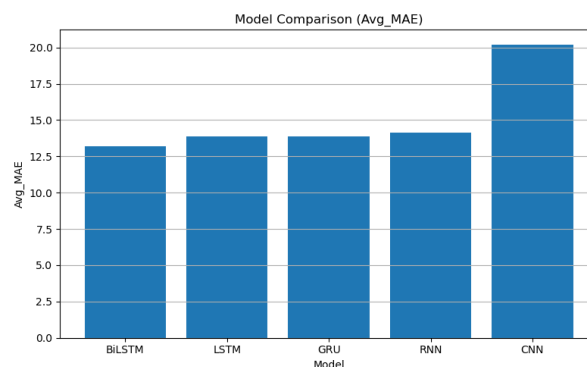


Figure 15: Comparison of Models Based on Average MAE

The MAE comparison shows that BiLSTM achieves the lowest error, indicating the highest prediction accuracy. LSTM and GRU follow closely with slightly higher MAE values, while RNN

performs moderately. CNN exhibits the highest MAE, highlighting its limited effectiveness for time-series forecasting Figure 15.

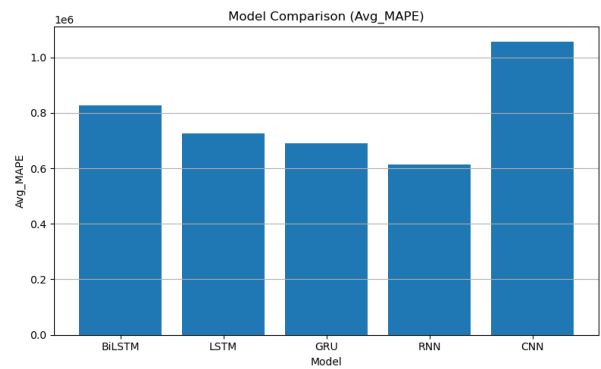


Figure 16: Comparison of Models Based on Average MAPE

The MAPE results indicate that RNN achieves the lowest percentage error, making it more consistent in relative prediction accuracy. GRU and LSTM also perform well, whereas BiLSTM shows slightly higher percentage error. CNN again records the highest MAPE, confirming poor performance in capturing trends Figure 16.

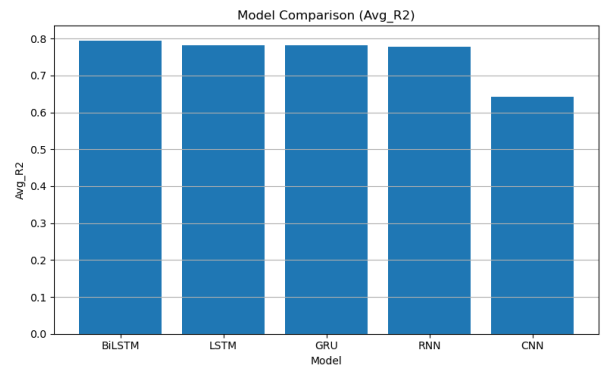


Figure 17: Comparison of Models Based on R^2 Score

The R^2 comparison highlights that BiLSTM achieves the highest score, indicating superior capability in explaining variance in the data. LSTM, GRU, and RNN show comparable performance, while CNN records the lowest R^2 , indicating weaker predictive power Figure 17.

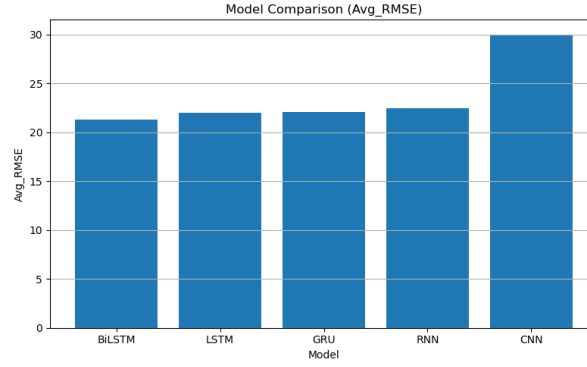


Figure 18: Comparison of Models Based on Average RMSE

The RMSE comparison shows that BiLSTM has the lowest error, confirming its superior performance in minimizing prediction deviations. LSTM and GRU perform similarly with slightly higher RMSE values. RNN shows moderate performance, while CNN has the highest RMSE, indicating poor prediction accuracy. BiLSTM achieves the best performance across all metrics, followed by LSTM and GRU. RNN shows moderate results, while CNN performs the worst, indicating its limitation in time-series forecasting Figure 18.

Hybrid Deep Learning Models Comparison of Chandni Chowk and RK Puram

7.1.7 Overall Hybrid Model Ranking Based on Average Metrics of Chandni Chowk

Table 9: Ranking of Hybrid Models Based on Average Evaluation Metrics

Rank	Model	Avg MSE	Avg RMSE	Avg MAE	Avg MAPE	R^2
1	Hybrid_BiLSTM_GRU	953.08	23.11	13.36	3160976.82	0.661
2	Hybrid_GRU_RNN	967.67	23.14	13.53	2666740.63	0.674
3	Hybrid_BiLSTM_CNN	1022.44	23.78	14.54	3064089.95	0.654
4	Hybrid_LSTM_CNN	1267.67	26.25	16.92	3032321.08	0.600
5	Hybrid_CNN_LSTM	1675.42	30.87	20.69	6196449.17	0.377

The Table 9 presents the overall performance comparison of hybrid deep learning models for air quality prediction at the Chandni Chowk location. Among all models, the Hybrid_BiLSTM_GRU model achieved the top rank with the lowest average RMSE (23.11) and MAE (13.36), indicating strong predictive accuracy. This demonstrates that combining bidirectional learning with gated recurrent units enhances the model's ability to capture complex temporal dependencies in highly dynamic environments. The Hybrid_GRU_RNN model also performed competitively, achieving the highest R^2 value (0.674), which indicates better variance explanation compared to other models. Its performance is closely aligned with the top-ranked model. Hybrid_BiLSTM_CNN showed moderate performance, suggesting that the combination of sequential and spatial feature extraction provides some improvement, but not as effective as purely recurrent hybrid architectures. The Hybrid_LSTM_CNN model resulted in higher error

values and lower R^2 , indicating limited effectiveness in capturing the complex temporal variations present in the Chandni Chowk dataset. The Hybrid_CNN_LSTM model performed the worst among all models, with the highest RMSE (30.87), MAE (20.69), and lowest R^2 value (0.377), suggesting that CNN-based hybrid structures are less suitable for this type of time-series prediction. Overall, it is evident that hybrid models combining recurrent architectures, particularly BiLSTM and GRU, outperform CNN-based hybrids for air quality prediction in highly congested urban environments like Chandni Chowk. Additionally, the extremely high MAPE values across all models indicate potential data scaling issues or the presence of near-zero actual values, making MAPE a less reliable evaluation metric in this context.

7.1.8 Overall Hybrid Model Ranking Based on Average Metrics of RK Puram

The Table 10 presents the overall performance comparison of hybrid deep learning models for air quality prediction at the RK Puram location. Among all models, the Hybrid_BiLSTM_GRU model achieved the best performance with the lowest average RMSE (23.58) and MAE (15.06), along with the highest R^2 value (0.862), indicating superior predictive accuracy. This demonstrates the effectiveness of combining bidirectional learning with GRU units for capturing temporal dependencies in relatively stable environments. The Hybrid_GRU_RNN model also showed strong performance, achieving an R^2 value of 0.855 with slightly higher error metrics. Its results are comparable to the top-performing model. Hybrid_LSTM_CNN and Hybrid_BiLSTM_CNN exhibited moderate performance, indicating that the inclusion of convolutional layers provides some benefit in feature extraction but is less effective compared to purely recurrent hybrid architectures. The Hybrid_CNN_LSTM model performed the worst among all models, with the highest RMSE (37.80), MAE (26.23), and lowest R^2 value (0.604), suggesting that CNN-dominant architectures are less suitable for time-series air quality prediction.

Table 10: Ranking of Hybrid Models Based on Average Evaluation Metrics

Rank	Model	Avg MSE	Avg RMSE	Avg MAE	Avg MAPE	R^2
1	Hybrid_BiLSTM_GRU	964.78	23.58	15.06	41.73	0.862
2	Hybrid_GRU_RNN	981.15	23.97	15.42	43.88	0.855
3	Hybrid_LSTM_CNN	1061.90	24.70	16.67	48.49	0.852
4	Hybrid_BiLSTM_CNN	1082.19	24.81	16.11	47.25	0.852
5	Hybrid_CNN_LSTM	2294.66	37.80	26.23	76.10	0.604

Overall, it is evident that hybrid models combining recurrent architectures, particularly BiLSTM and GRU, outperform CNN-based hybrids for air quality prediction in RK Puram. Compared to Chandni Chowk, the models achieve higher R^2 values and lower error metrics, indicating that the relatively stable environmental conditions of RK Puram allow better learning and generalization. Additionally, unlike the Chandni Chowk dataset, the MAPE values are within a reasonable range, making it a more reliable evaluation metric for this location.

7.1.9 Featurewise Hybrid Models Performance Across Pollutants of Chandni Chowk

The Table 11 presents the feature-wise performance comparison of hybrid deep learning models for different air pollutants at the Chandni Chowk location. For PM_{2.5} prediction, Hybrid BiLSTM-GRU achieved the best performance with the lowest RMSE (35.10) and highest R^2 value (0.596), closely followed by Hybrid GRU-RNN. CNN-based hybrids, particularly Hybrid CNN-LSTM, showed significantly higher error values and poor predictive capability. In the case of PM₁₀, Hybrid BiLSTM-GRU again performed best with the highest R^2 value (0.809) and lowest RMSE, followed closely by Hybrid GRU-RNN. Hybrid CNN-LSTM exhibited the weakest performance. For NO₂, Hybrid LSTM-CNN achieved the highest R^2 value (0.781), indicating slightly better performance compared to other models. Hybrid GRU-RNN also showed competitive results, while CNN-LSTM again underperformed. For CO prediction, Hybrid GRU-RNN demonstrated the best performance with the highest R^2 value (0.547) and lowest RMSE. However, extremely high MAPE values across all models indicate the presence of near-zero actual values, making MAPE less reliable for this pollutant. In the case of Ozone, Hybrid GRU-RNN achieved the best performance with the highest R^2 value (0.651), followed by Hybrid BiLSTM-CNN and BiLSTM-GRU. CNN-based hybrids again resulted in significantly higher error values.

Overall, hybrid models combining recurrent architectures such as BiLSTM and GRU consistently outperform CNN-based hybrids for most pollutants. Models incorporating CNN components tend to show weaker performance, particularly in highly dynamic and noisy environments like Chandni Chowk. The results indicate that temporal dependencies play a dominant role in air quality prediction, and models focusing on sequential learning are more effective for this application.

7.1.10 Featurewise Models Performance Across Environmental Parameters of RK Puram

The Table 12 presents the feature-wise performance comparison of hybrid deep learning models for different air pollutants at the RK Puram location. For PM_{2.5} prediction, Hybrid BiLSTM-GRU achieved the best performance with the lowest RMSE (37.59) and highest R^2 value (0.869), followed by Hybrid GRU-RNN. Hybrid CNN-LSTM showed significantly higher error values and lower predictive accuracy. In the case of PM₁₀, Hybrid GRU-RNN slightly outperformed other models with the highest R^2 value (0.833), followed closely by Hybrid BiLSTM-GRU. CNN-LSTM again resulted in the weakest performance. For NO₂, Hybrid LSTM-CNN and Hybrid BiLSTM-CNN achieved the highest R^2 values (0.876), indicating strong performance. Hybrid GRU-RNN and BiLSTM-GRU also demonstrated competitive results, while CNN-LSTM remained the least effective. For CO prediction, Hybrid BiLSTM-GRU achieved the best performance with the highest R^2 value (0.842) and lowest RMSE. Other recurrent hybrids also performed well, while CNN-LSTM showed significantly poor results. In the case of

Table 11: Hybrid Model Performance Across Different Air Pollutants of Chandni Chowk

Pollutant	Hybrid LSTM-CNN				Hybrid BiLSTM-GRU				Hybrid GRU-RNN				Hybrid CNN-LSTM				Hybrid BiLSTM-CNN			
	RMSE	MAE	MAPE	R ²	RMSE	MAE	MAPE	R ²	RMSE	MAE	MAPE	R ²	RMSE	MAE	MAPE	R ²	RMSE	MAE	MAPE	R ²
PM2.5	44.11	31.86	65.37	0.361	35.10	23.18	37.16	0.596	35.29	23.03	37.19	0.591	46.40	34.81	72.89	0.293	36.72	25.01	46.56	0.557
PM10	64.23	40.15	28.70	0.758	57.09	31.40	23.97	0.809	57.73	32.94	26.69	0.805	75.41	49.08	35.91	0.667	59.13	34.84	26.25	0.795
NO ₂	8.71	6.27	18.68	0.781	9.26	6.65	17.89	0.752	8.82	6.27	18.45	0.775	12.67	8.91	21.85	0.536	9.08	6.49	18.12	0.762
CO	0.379	0.237	15161456.16	0.468	0.363	0.256	15804778.67	0.511	0.349	0.226	13333598.31	0.547	0.489	0.385	30982046.19	0.113	0.363	0.235	15320323.48	0.512
Ozone	13.82	6.09	36.48	0.633	13.72	5.34	26.43	0.639	13.48	5.17	22.51	0.651	19.40	10.27	69.00	0.278	13.58	6.13	35.33	0.646

Ozone prediction, Hybrid BiLSTM-GRU demonstrated the best performance with the highest R^2 value (0.894), followed closely by Hybrid BiLSTM-CNN and LSTM-CNN.

Overall, hybrid models combining recurrent architectures such as BiLSTM, GRU, and RNN consistently outperform CNN-based hybrids. The results for RK Puram show higher R^2 values and lower error metrics compared to Chandni Chowk, indicating that relatively stable environmental conditions allow models to learn more effectively.

7.1.11 Hybrid Model Comparison of Chandni Chowk

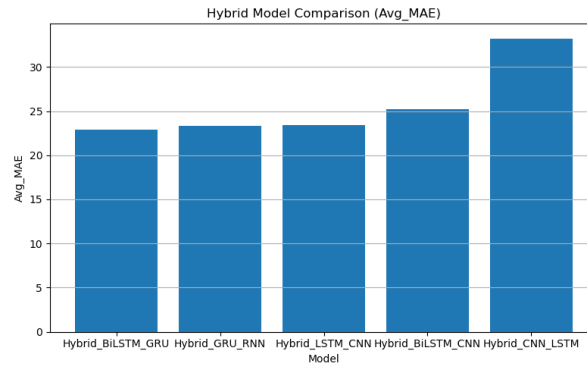


Figure 19: Hybrid Model Comparison based on Average MAE

The comparison of hybrid models based on average MAE shows that the Hybrid_BiLSTM_GRU model achieves the lowest error, indicating superior predictive performance. Hybrid_GRU_RNN and Hybrid_LSTM_CNN also perform competitively with slightly higher MAE values. In contrast, Hybrid_CNN_LSTM exhibits the highest MAE, suggesting relatively poorer accuracy Figure 19.

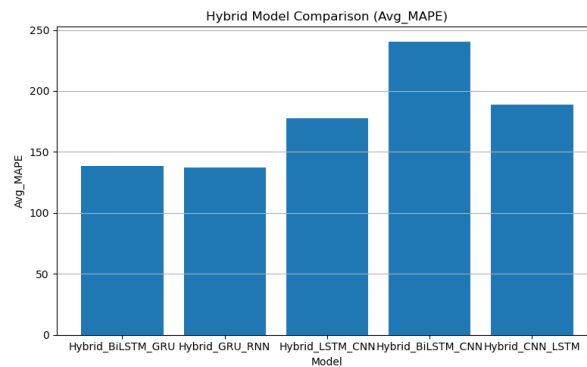


Figure 20: Hybrid Model Comparison based on MAPE

In terms of MAPE, Hybrid_GRU_RNN and Hybrid_BiLSTM_GRU demonstrate lower percentage errors, indicating better consistency in predictions. Hybrid_BiLSTM_CNN shows moderate performance, while Hybrid_CNN_LSTM again performs the worst with significantly higher MAPE values Figure 20.

Table 12: Hybrid Model Performance Across Different Air Pollutants of RK Puram

Pollutant	Hybrid LSTM-CNN				Hybrid BiLSTM-GRU				Hybrid GRU-RNN				Hybrid CNN-LSTM				Hybrid BiLSTM-CNN			
	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE	R^2
PM2.5	41.49	26.50	20.14	0.840	37.59	22.82	17.40	0.869	38.77	23.62	17.70	0.860	64.42	41.56	28.59	0.615	40.94	25.32	20.26	0.844
PM10	57.38	39.79	22.97	0.822	55.86	35.85	19.62	0.831	55.53	36.29	20.84	0.833	79.56	56.84	31.58	0.658	58.69	38.62	21.76	0.814
NO ₂	10.47	7.81	21.23	0.876	10.60	7.46	17.88	0.873	10.85	7.42	17.64	0.867	19.19	14.12	34.67	0.584	10.49	7.32	16.93	0.876
CO	0.522	0.332	65.55	0.833	0.508	0.299	52.76	0.842	0.525	0.310	53.32	0.831	0.874	0.586	96.38	0.532	0.523	0.322	58.51	0.832
Ozone	13.65	8.90	112.58	0.890	13.35	8.86	101.01	0.894	14.17	9.48	109.90	0.881	24.98	18.02	189.28	0.630	13.42	9.00	118.81	0.893

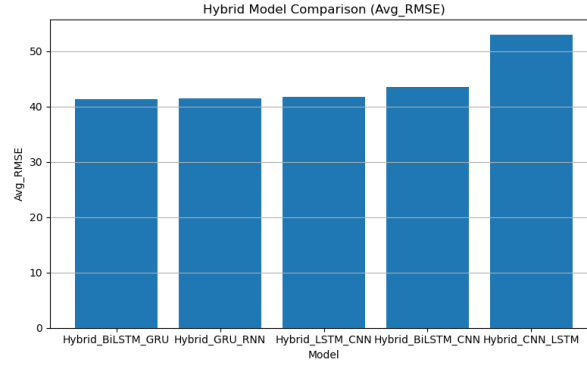


Figure 21: Hybrid Model Comparison based on RMSE

The RMSE comparison reveals that Hybrid_BiLSTM_GRU achieves the lowest root mean square error, indicating high prediction accuracy. Hybrid_GRU_RNN and Hybrid_LSTM_CNN follow closely, while Hybrid_CNN_LSTM shows the highest RMSE, reflecting larger prediction deviations Figure 21.

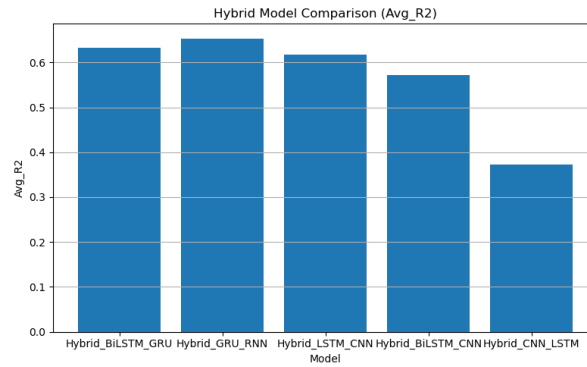


Figure 22: Hybrid Model Comparison based on R^2 Score

The R^2 comparison indicates that Hybrid_BiLSTM_GRU achieves the highest coefficient of determination, suggesting better model fit and predictive capability. Hybrid_GRU_RNN also performs well, whereas Hybrid_CNN_LSTM has the lowest R^2 , indicating weaker explanatory power Figure 22.

7.1.12 Hybrid Model Comparison of RK Puram

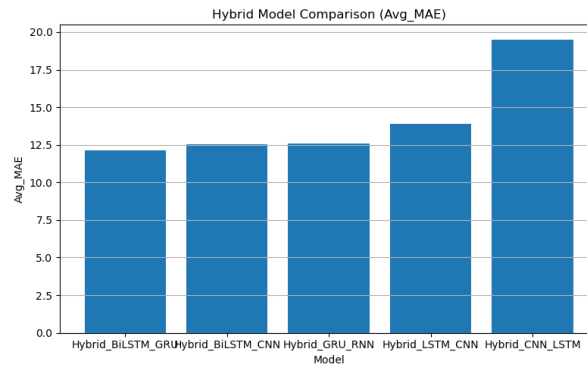


Figure 23: Hybrid Model Comparison Based on Average MAE

The MAE comparison shows that the Hybrid_BiLSTM_GRU model achieves the lowest error (approximately 12.2), indicating the highest prediction accuracy among all models. Hybrid_BiLSTM_CNN and Hybrid_GRU_RNN follow closely with slightly higher MAE values, demonstrating stable and reliable performance. Hybrid_LSTM_CNN shows moderate performance, while Hybrid_CNN_LSTM records the highest MAE (around 19.5), indicating poor predictive accuracy Figure 23.

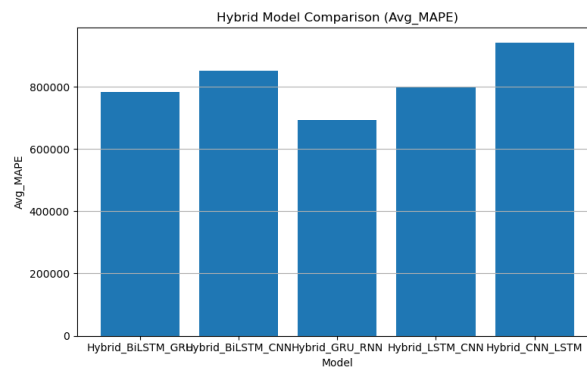


Figure 24: Hybrid Model Comparison Based on MAPE

Based on MAPE, Hybrid_GRU_RNN achieves the lowest percentage error (around 690,000), followed by Hybrid_BiLSTM_GRU, indicating strong consistency in prediction. Hybrid_BiLSTM_CNN and Hybrid_LSTM_CNN show moderate error levels. Hybrid_CNN_LSTM again performs the worst with the highest MAPE (around 930,000), confirming its lower reliability in relative error measurement Figure 24.

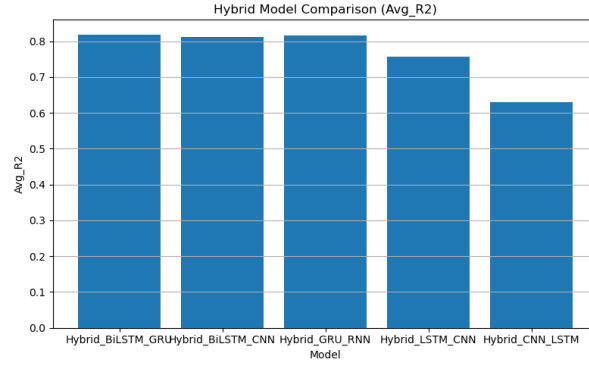


Figure 25: Hybrid Model Comparison Based on R^2 Score

The R^2 score comparison indicates that Hybrid_BiLSTM_GRU and Hybrid_GRU_RNN achieve the highest values (approximately 0.82), demonstrating strong model fit and explanatory power. Hybrid_BiLSTM_CNN also performs well with a slightly lower value. Hybrid_LSTM_CNN shows moderate performance, while Hybrid_CNN_LSTM has the lowest R^2 (around 0.63), indicating weak predictive capability Figure 25.

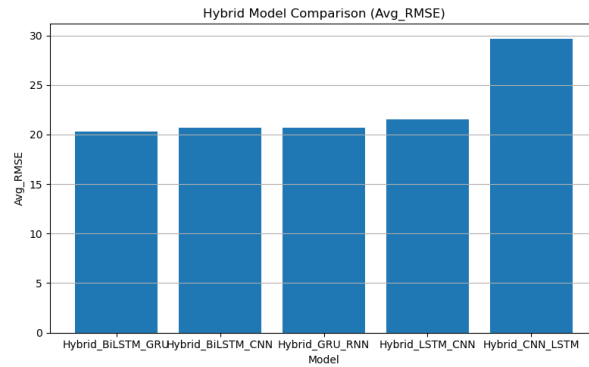


Figure 26: Hybrid Model Comparison Based on RMSE

The RMSE results reveal that Hybrid_BiLSTM_GRU achieves the lowest error (approximately 20.2), indicating minimal prediction deviation. Hybrid_BiLSTM_CNN and Hybrid_GRU_RNN show similar performance with slightly higher RMSE values. Hybrid_LSTM_CNN performs moderately, whereas Hybrid_CNN_LSTM again records the highest RMSE (around 29.7), reflecting significant prediction errors. Considering all evaluation metrics (MAE, MAPE, RMSE, and R^2), the Hybrid_BiLSTM_GRU model consistently outperforms the other hybrid models, making it the most effective for time-series prediction. Hybrid_GRU_RNN and Hybrid_BiLSTM_CNN also demonstrate strong and stable performance. In contrast, Hybrid_CNN_LSTM consistently ranks the lowest across all metrics, highlighting its limitations in capturing temporal dependencies effectively Figure 26.

7.2 Quantitative Analysis

The quantitative analysis for RK Puram indicates that the Hybrid BiLSTM-GRU model outperforms all conventional models including RNN, LSTM, GRU, CNN, and BiLSTM. Significant percentage improvement is observed for particulate matter (PM_{2.5} and PM₁₀) due to effective temporal feature extraction. Moderate improvement is achieved for NO₂ and CO, while comparatively lower gains are seen for O₃ due to its complex atmospheric behavior. The hybrid model consistently reduces error metrics such as MSE, RMSE, and MAE across all pollutants. Overall, the proposed model demonstrates superior prediction accuracy and robustness for real-time air quality forecasting.

7.2.1 Quantitative Analysis Based on MAE and Percentage Improvement

To evaluate the effectiveness of the proposed Hybrid BiLSTM-GRU model, a comparative analysis is performed using Mean Absolute Error (MAE) across multiple pollutants. Additionally, percentage improvement is calculated to quantify the performance gain of the hybrid model over individual deep learning models.

Table 13: MAE and Percentage Improvement of Hybrid Model over Other Models

Pollutant	Hybrid BiLSTM-GRU	BiLSTM	CNN	GRU	LSTM	RNN
PM _{2.5}	22.8195	25.1461 (9.25%)	38.9746 (41.45%)	26.8777 (15.1%)	25.0536 (8.92%)	24.8828 (8.29%)
PM ₁₀	35.8536	37.3933 (4.12%)	54.6127 (34.35%)	37.2837 (3.84%)	36.2059 (0.97%)	36.5281 (1.85%)
NO ₂	7.4554	7.7530 (3.84%)	12.2061 (38.92%)	8.1138 (8.12%)	7.7596 (3.92%)	7.7997 (4.41%)
CO	0.2989	0.3632 (17.71%)	0.5722 (47.76%)	0.3722 (19.68%)	0.3582 (16.56%)	0.3620 (17.42%)
Ozone	8.8645	9.2826 (4.5%)	15.7147 (43.59%)	9.8737 (10.22%)	9.4725 (6.42%)	10.2905 (13.86%)

The Table 13 quantitative analysis based on MAE and percentage improvement clearly demonstrates the superiority of the Hybrid BiLSTM-GRU model over all standalone deep learning models. The hybrid model consistently achieves the lowest MAE values across all pollutants, indicating higher prediction accuracy and better generalization capability. For particulate matter (PM_{2.5} and PM₁₀), which are highly dynamic and influenced by multiple emission sources, the hybrid model shows significant improvements. In PM_{2.5} prediction, it achieves a 41.45% improvement over CNN and notable gains over GRU (15.1%), BiLSTM (9.25%), LSTM (8.92%), and RNN (8.29%). This highlights the hybrid model's ability to effectively capture complex temporal dependencies and variability in particulate pollution.

In the case of PM₁₀, although improvements over sequential models such as LSTM and RNN are relatively smaller (0.97% and 1.85%), a substantial improvement of 34.35% is observed over CNN, reinforcing the limitation of convolution-based architectures in handling time-series air quality data. For gaseous pollutants like NO₂ and CO, the hybrid model continues to outperform all models. In CO prediction, the hybrid model achieves the highest improvement of 47.76% over CNN and around 17–19% over other recurrent models. This indicates its strong capability in capturing subtle variations and low-magnitude pollutant concentrations.

Similarly, for NO₂, the hybrid model achieves consistent improvements across all models, particularly a significant 38.92% improvement over CNN. For Ozone, which involves complex atmospheric chemistry and nonlinear interactions, the hybrid model again shows strong performance with a 43.59% improvement over CNN and notable gains over GRU and RNN. The superior performance of the Hybrid BiLSTM-GRU model can be attributed to its architectural design, which combines the strengths of BiLSTM and GRU. The BiLSTM component captures bidirectional temporal dependencies, allowing the model to learn from both past and future context, while the GRU component efficiently controls information flow through gating mechanisms, reducing computational complexity and improving learning efficiency.

Overall, the results confirm that the hybrid approach significantly enhances prediction accuracy compared to individual models. It effectively addresses the limitations of standalone architectures and proves to be highly suitable for multi-pollutant air quality forecasting tasks.

7.2.2 Quantitative Analysis Based on MAPE and Percentage Improvement

To further evaluate model performance in terms of relative error, Mean Absolute Percentage Error (MAPE) is used. MAPE provides a normalized measure of prediction accuracy, making it suitable for comparing pollutants with different scales. Lower MAPE values indicate better predictive performance.

Table 14: MAPE and Percentage Improvement of Hybrid Model over Other Models

Pollutant	Hybrid BiLSTM-GRU	BiLSTM	CNN	GRU	LSTM	RNN
PM _{2.5}	1412.98	1817.38 (22.25%)	3880.55 (63.59%)	2137.86 (33.91%)	1854.19 (23.8%)	1987.38 (28.9%)
PM ₁₀	3120.06	3315.27 (5.89%)	6554.07 (52.4%)	3545.68 (12%)	3295.52 (5.32%)	3270.52 (4.6%)
NO ₂	112.41	132.63 (15.24%)	294.21 (61.79%)	156.72 (28.27%)	134.09 (16.16%)	138.91 (19.08%)
CO	0.2579	0.3063 (15.8%)	0.7092 (63.63%)	0.3482 (25.94%)	0.3117 (17.25%)	0.3274 (21.22%)
Ozone	178.20	261.10 (31.75%)	605.14 (70.55%)	282.25 (36.87%)	267.48 (33.38%)	305.10 (41.59%)

The MAPE-based analysis provides deeper insights into the relative prediction accuracy of the models across pollutants with varying concentration ranges. The results clearly indicate that the Hybrid BiLSTM-GRU model achieves the lowest MAPE values for all pollutants, confirming its superior performance in minimizing relative prediction errors. The Table 14. For PM_{2.5}, which exhibits high variability and non-linear behavior, the hybrid model significantly reduces prediction error, achieving a 63.59% improvement over CNN and more than 22% improvement over BiLSTM and LSTM. This demonstrates its ability to effectively capture complex temporal dependencies and fluctuations in particulate matter concentrations. In PM₁₀ prediction, the hybrid model shows moderate improvements over recurrent models (approximately 4–12%) but achieves a substantial 52.4% improvement over CNN. This indicates that while sequential models perform reasonably well, the hybrid architecture provides additional refinement in prediction accuracy. For NO₂, the hybrid model again outperforms all models, achieving up to 61.79% improvement over CNN and consistent improvements ranging from 15% to 28% over

other recurrent models. This highlights the model’s robustness in handling relatively stable gaseous pollutants. In the case of CO, the hybrid model demonstrates strong performance with a 63.63% improvement over CNN and around 15–25% over other models. Since CO concentrations are relatively low and sensitive to minor variations, the reduced MAPE indicates improved precision in capturing subtle changes.

Ozone prediction shows the highest overall improvement, with the hybrid model achieving a remarkable 70.55% improvement over CNN and more than 30% improvement over all recurrent models. This is particularly significant because ozone formation involves complex photochemical reactions and nonlinear dependencies. The superior performance of the Hybrid BiLSTM-GRU model can be attributed to its combined architecture. The BiLSTM component captures bidirectional temporal patterns, while the GRU component efficiently regulates information flow using gating mechanisms. This combination enhances the model’s ability to generalize across different pollutants and varying concentration levels.

Overall, the MAPE analysis confirms that the hybrid model not only reduces absolute errors but also significantly improves relative prediction accuracy. This makes it highly reliable for real-world air quality forecasting, where accurate percentage-based predictions are critical for decision-making and environmental management.

7.2.3 Quantitative Analysis Based on RMSE and Percentage Improvement

Root Mean Square Error (RMSE) is used to evaluate model performance by penalizing larger prediction errors more heavily. It provides insight into how well the model handles significant deviations between predicted and actual values.

Table 15: RMSE and Percentage Improvement of Hybrid Model over Other Models

Pollutant	Hybrid BiLSTM-GRU	BiLSTM	CNN	GRU	LSTM	RNN
PM _{2.5}	37.5896	42.6307 (11.83%)	62.2941 (39.66%)	46.2370 (18.7%)	43.0603 (12.7%)	44.5800 (15.68%)
PM ₁₀	55.8575	57.5784 (2.99%)	80.9572 (31%)	59.5456 (6.19%)	57.4066 (2.7%)	57.1885 (2.33%)
NO ₂	10.6024	11.5163 (7.94%)	17.1527 (38.19%)	12.5190 (15.31%)	11.5796 (8.44%)	11.7862 (10.04%)
CO	0.5078	0.5534 (8.24%)	0.8421 (39.7%)	0.5901 (13.94%)	0.5583 (9.03%)	0.5721 (11.24%)
Ozone	13.3491	16.1586 (17.39%)	24.5997 (45.73%)	16.8004 (20.54%)	16.3550 (18.38%)	17.4671 (23.58%)

The RMSE-based evaluation highlights the ability of models to handle large prediction errors, making it a critical metric for assessing robustness in air quality forecasting. The results clearly indicate that the Hybrid BiLSTM-GRU model achieves the lowest RMSE values across all pollutants, demonstrating superior predictive stability and accuracy.

For PM_{2.5}, which is highly volatile and influenced by diverse emission sources, the hybrid model significantly reduces large prediction errors. It achieves a 39.66% improvement over CNN and notable improvements over GRU (18.7%), LSTM (12.7%), BiLSTM (11.83%), and RNN (15.68%). This confirms its ability to capture extreme variations in particulate matter concentrations the Table 15. In PM₁₀ prediction, improvements over recurrent models are relatively

moderate (approximately 2–6%), indicating that sequential models already perform reasonably well. However, the hybrid model still achieves a substantial 31% improvement over CNN, highlighting its superiority in handling temporal dependencies compared to convolution-based approaches. For NO_2 , the hybrid model demonstrates consistent performance gains, achieving up to 38.19% improvement over CNN and 7–15% improvements over recurrent models. This reflects its robustness in modeling relatively stable gaseous pollutants with occasional fluctuations. In the case of CO, the hybrid model reduces RMSE by approximately 39.7% compared to CNN and around 8–14% compared to other models. Since RMSE penalizes larger errors more heavily, this reduction indicates improved reliability in predicting sudden deviations in CO concentrations. Ozone prediction shows strong improvements, with a 45.73% reduction in RMSE compared to CNN and 17–23% improvements over recurrent models. Given the nonlinear and photochemical nature of ozone formation, this highlights the hybrid model's capability to effectively capture complex atmospheric interactions.

The superior performance of the Hybrid BiLSTM-GRU model can be attributed to the integration of BiLSTM and GRU architectures. The BiLSTM component captures bidirectional temporal relationships, while the GRU component enhances learning efficiency through simplified gating mechanisms. This combination enables the model to minimize both average and extreme prediction errors.

Overall, the RMSE analysis confirms that the hybrid model is highly effective in reducing large prediction deviations, making it more robust and reliable for real-world air quality forecasting applications.

7.3 Graphical Analysis

Graphical analysis is carried out to visually interpret the performance of different deep learning models for air quality prediction across two major locations in Delhi, namely Chandni Chowk and R.K. Puram. In this study, line plots are used to compare the actual and predicted values of $\text{PM}_{2.5}$ concentrations. The models evaluated include GRU, LSTM, BiLSTM, RNN, and CNN. These graphical representations help in understanding how well each model captures the temporal variations and trends in air pollution levels.

7.3.1 Predictive Pattern Analysis

The predictive pattern analysis for Chandni Chowk is conducted using recent hourly data spanning consecutive days. The dataset includes $\text{PM}_{2.5}$ along with pollutants such as PM_{10} , NO_2 , CO, and O_3 . The predictive pattern analysis for R.K. Puram is carried out using a time-series line plot based on recent hourly air quality data spanning consecutive days (e.g., 02-14 to 02-15). The dataset includes $\text{PM}_{2.5}$ concentration levels along with other major air pollutants such as PM_{10} , NO_2 , CO, and O_3 . The line plot illustrates the comparison between actual $\text{PM}_{2.5}$ values and the predicted outputs generated by different deep learning models, namely GRU,

LSTM, BiLSTM, RNN, and CNN, over multiple time intervals. From the graphical representation, it is observed that the GRU model closely follows the actual trend with minimal deviation, demonstrating strong predictive performance. The LSTM and BiLSTM models are able to capture the overall pattern but show slight discrepancies, particularly during peak pollution periods. The RNN model shows moderate accuracy, while the CNN model exhibits significant deviation from actual values and fails to capture sharp fluctuations in $PM_{2.5}$ concentration. The line plot compares actual $PM_{2.5}$ values with predicted outputs from GRU, LSTM, BiLSTM, RNN, and CNN models.

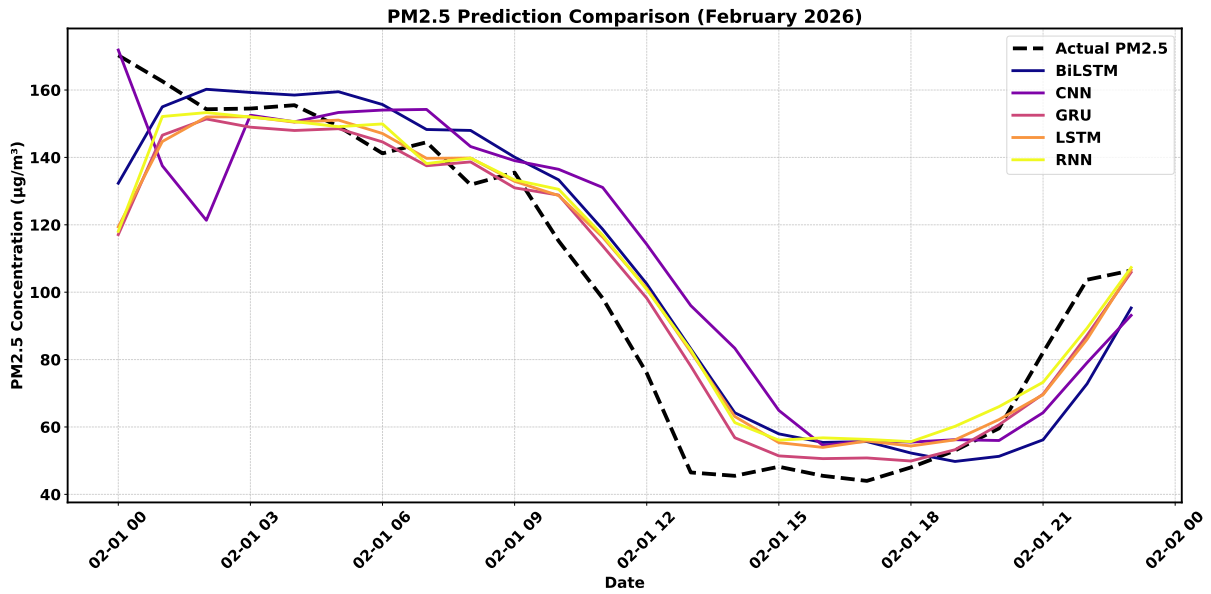


Figure 27: $PM_{2.5}$ Prediction Comparison using Deep Learning Models(Chandni Chowk)

From Figure 27. it is observed that the GRU model closely follows the actual trend with minimal deviation, indicating high prediction accuracy. LSTM and BiLSTM capture the general pattern with slight variations, while RNN shows moderate performance. CNN exhibits the largest deviation and fails to capture sudden fluctuations effectively. Overall, the graphical analysis confirms that GRU provides the most accurate predictions for Chandni Chowk, consistent with the quantitative analysis results.

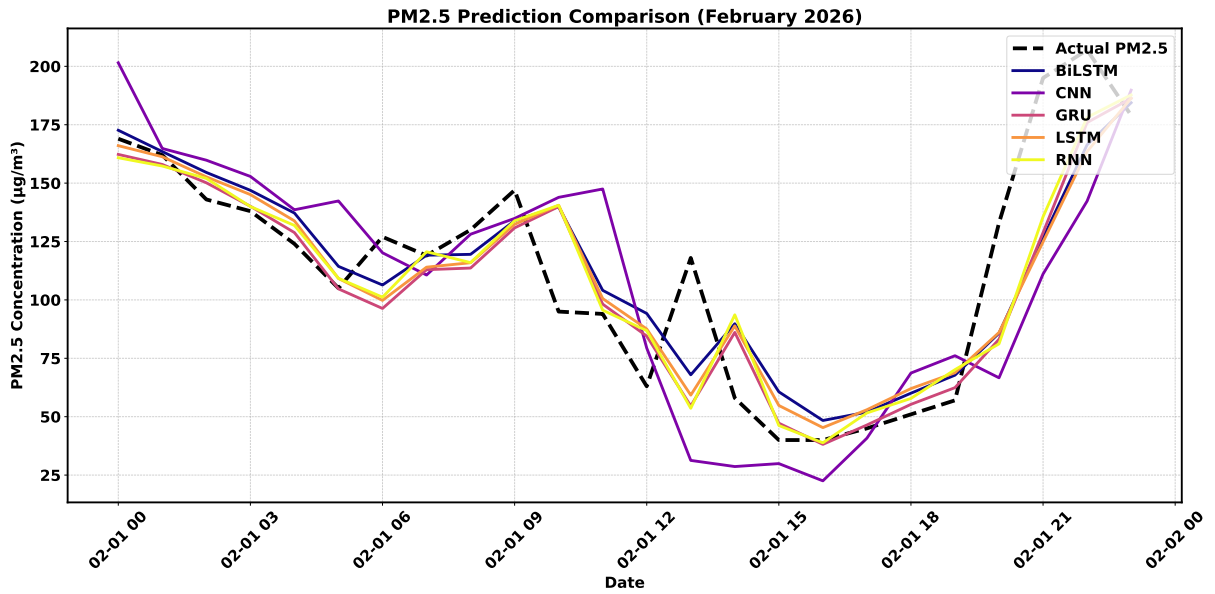


Figure 28: PM_{2.5} Prediction Comparison using Deep Learning Models(RK Puram)

From Figure 28, it is evident that GRU provides the most accurate predictions for R.K. Puram, followed by LSTM and BiLSTM. The CNN model performs the worst due to its inability to effectively model temporal dependencies in the dataset. The graphical findings are consistent with the quantitative analysis, further confirming the robustness and reliability of the GRU model for air quality prediction at R.K. Puram.

7.3.2 Prediction Distributive Analysis

The prediction distribution analysis is performed using superimposed scatter plots of predicted versus actual PM_{2.5} concentrations for different deep learning models at both Chandni Chowk and RK Puram. These plots provide a visual assessment of model accuracy, dispersion, and correlation by comparing the alignment of data points along the 45-degree reference line. For Chandni Chowk, the scatter plots indicate that the GRU and RNN models exhibit a relatively strong linear alignment with the ideal diagonal line, suggesting better predictive capability and higher correlation values. The GRU model demonstrates the most compact clustering of points around the reference line, indicating lower prediction error and better generalization. The RNN model follows closely, although with slightly higher dispersion. In contrast, the LSTM and BiLSTM models show moderate spread, indicating increased variability in predictions. The CNN model displays the highest dispersion and noticeable deviation from the diagonal line, particularly at higher PM_{2.5} concentrations, reflecting its limited effectiveness in capturing temporal dependencies in highly fluctuating pollution patterns. For RK Puram, the scatter plots reveal an overall improvement in prediction performance across all models compared to Chandni Chowk, as indicated by tighter clustering and higher correlation values. The GRU, LSTM, and BiLSTM models demonstrate strong alignment with the reference line, indicating high predictive accuracy and consistency. Among these, the GRU model again shows the most concentrated dis-

tribution of points, confirming its robustness. The RNN model also performs well but exhibits slightly greater dispersion compared to GRU and LSTM. The CNN model, although improved relative to its performance at Chandni Chowk, still shows wider spread and deviation, especially at extreme values, indicating its comparatively weaker performance.

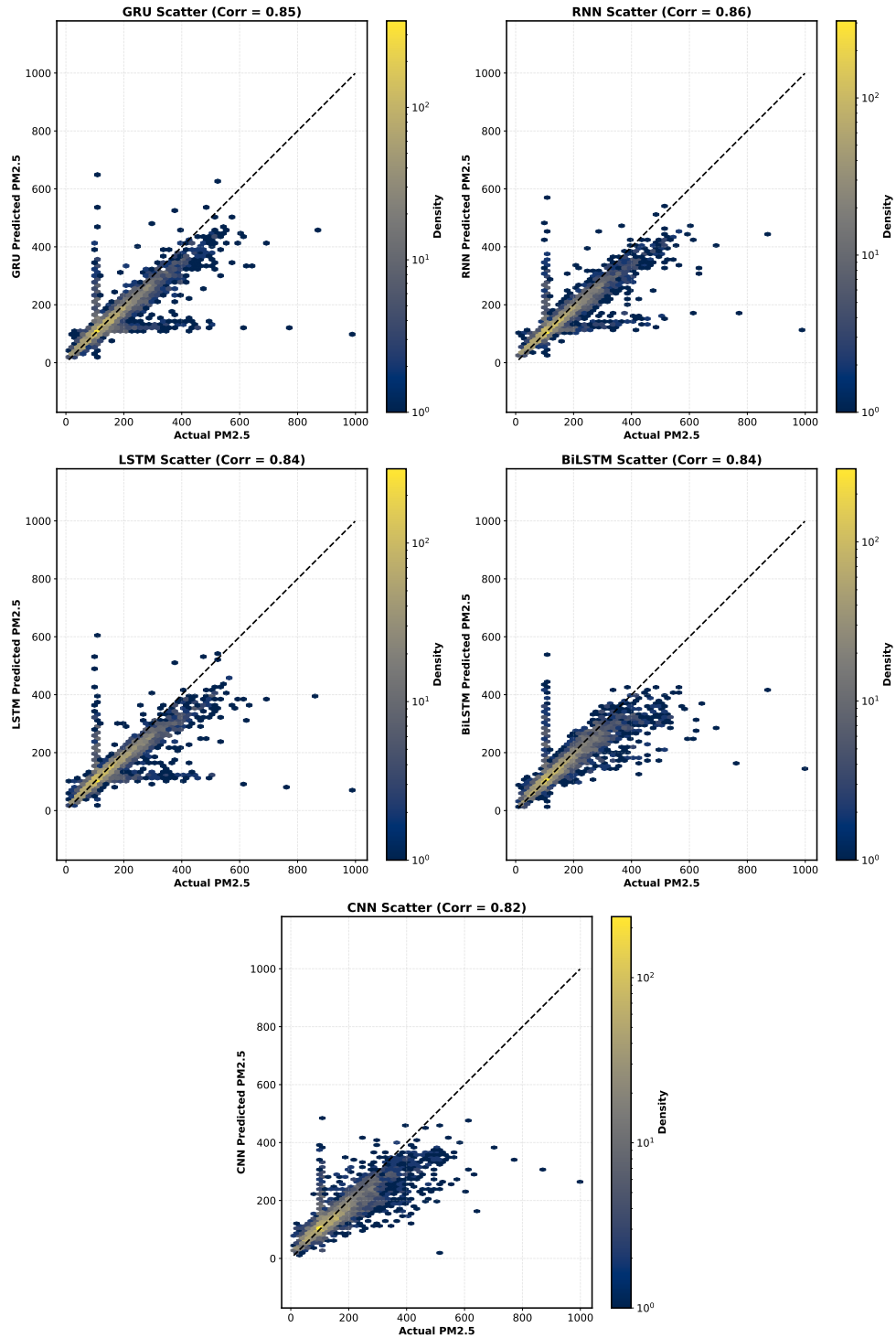


Figure 29: Superimposed Scatter Plots of Predicted vs Actual $PM_{2.5}$ for Chandni Chowk

From Figure 29. The scatter plots illustrate the predictive performance of five deep learning

models: GRU, RNN, LSTM, BiLSTM, and CNN. Each plot compares predicted $PM_{2.5}$ concentrations against actual observed values.

GRU Model:The GRU model demonstrates a strong correlation (approximately 0.85) between predicted and actual values. The data points are closely clustered along the diagonal line, indicating high prediction accuracy and low variance. This suggests that GRU effectively captures temporal dependencies in air quality data.

RNN Model:The RNN model exhibits the highest correlation (approximately 0.86), showing a good alignment between predicted and actual values. However, slight dispersion in higher concentration ranges indicates some limitations in capturing extreme pollution events.

LSTM Model:The LSTM model achieves a correlation of around 0.84. The scatter points are moderately aligned with the diagonal, indicating stable performance. However, compared to GRU and RNN, it shows slightly higher spread, especially for peak values.

BiLSTM Model:The BiLSTM model also shows a correlation of approximately 0.84. The bidirectional structure helps capture both past and future dependencies, but the scatter indicates moderate dispersion, suggesting room for improvement in prediction consistency.

CNN Model:The CNN model demonstrates a correlation of around 0.82, which is slightly lower than other models. The scatter plot shows a wider spread of points, indicating less accurate predictions and weaker temporal learning capability compared to recurrent models.

Overall, the scatter plot analysis reveals that GRU and RNN models outperform others in capturing the relationship between predicted and actual $PM_{2.5}$ concentrations for Chandni Chowk. CNN shows comparatively weaker performance due to its limited ability to model temporal dependencies in time-series air quality data.

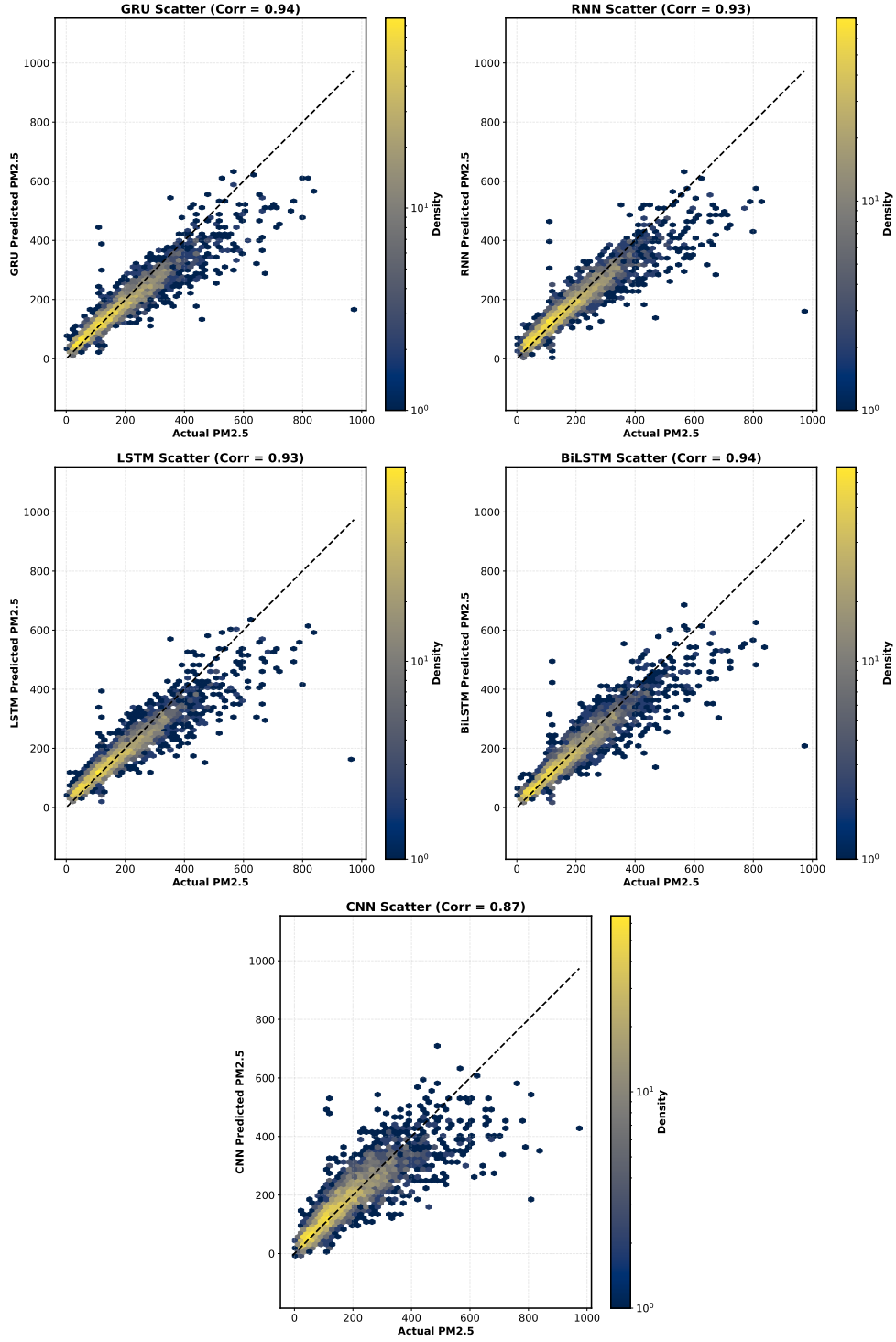


Figure 30: Superimposed Scatter Plots of Predicted vs Actual $PM_{2.5}$ for RK Puram

From Figure 30. The scatter plots represent the relationship between predicted and actual $PM_{2.5}$ concentrations for RK Puram using different deep learning models. A closer alignment of points along the diagonal line indicates better predictive accuracy.

GRU Model: The GRU model exhibits a very high correlation of approximately 0.94, indicating excellent agreement between predicted and actual values. The points are tightly clustered along the diagonal line, suggesting that the model effectively captures temporal depen-

dencies and performs robustly across varying pollution levels. **RNN Model:**The RNN model also demonstrates strong predictive capability with a correlation of around 0.93. The scatter points show good alignment, although slight dispersion is observed at higher PM_{2.5} concentrations, indicating minor limitations in modeling extreme pollution events. **LSTM Model:**The LSTM model achieves a correlation of approximately 0.93, reflecting stable and consistent performance. The scatter distribution is well-aligned with the diagonal, indicating that the model effectively learns long-term dependencies in time-series air quality data. **BiLSTM Model:**The BiLSTM model shows one of the best performances with a correlation of about 0.94. The bidirectional learning capability enhances its prediction accuracy, resulting in tightly clustered data points and minimal variance across the dataset. **CNN Model:**The CNN model demonstrates comparatively lower performance with a correlation of approximately 0.87. The scatter plot shows a wider spread of points, indicating less precise predictions. This is due to the limited ability of CNN to capture temporal dependencies in sequential air quality data. Overall, the results indicate that GRU and BiLSTM models provide the highest prediction accuracy for RK Puram, followed closely by LSTM and RNN models. The CNN model performs relatively weaker due to its architectural limitations in handling time-series data.

7.4 Statistical Analysis of Results Based on Measured Values and Prediction

The Non-Parametric Friedman Ranking test is employed to compare multiple models across repeated observations. This test ranks the models based on their predictive performance and helps identify statistically significant differences among GRU, LSTM, BiLSTM, RNN, and CNN models. It provides a reliable framework for determining the best-performing model under varying environmental conditions. To evaluate model complexity and avoid overfitting, Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are used. These criteria penalize excessive model parameters while rewarding goodness of fit, ensuring that selected models achieve an optimal balance between accuracy and simplicity. Lower AIC and BIC values indicate better model suitability for forecasting air quality. Furthermore, the Diebold-Mariano (DM) test is applied to compare the predictive accuracy between pairs of models. This statistical test determines whether the difference in forecast errors between two models is significant. It plays a crucial role in validating whether improvements in prediction performance are statistically meaningful rather than due to random variations. In addition, a Taylor Diagram is utilized to provide a comprehensive graphical summary of model performance. It simultaneously represents three key statistical measures: correlation coefficient, standard deviation, and root mean square error (RMSE). This integrated visualization allows for a direct comparison of model accuracy and variability relative to observed data. The combination of these statistical techniques provides a robust and comprehensive evaluation framework.

7.4.1 Taylor Diagram Results Analysis

The Taylor diagram provides a comprehensive graphical representation of model performance by simultaneously illustrating the correlation coefficient, standard deviation, and root mean square error (RMSE) between predicted and observed $\text{PM}_{2.5}$ concentrations. In the Chandni Chowk diagram, the GRU and BiLSTM models are positioned closer to the reference (observed data), indicating higher correlation and lower deviation, thus demonstrating superior predictive performance compared to LSTM, RNN, and CNN models. The CNN model is relatively farther from the reference point, reflecting lower correlation and higher variability. Similarly, in the RK Puram diagram, GRU and BiLSTM models again exhibit the best alignment with observed data, confirming their robustness across different urban environments. The clustering of GRU, BiLSTM, and LSTM near the reference arc indicates strong agreement with actual measurements, while CNN shows noticeable deviation due to its limited capability in capturing temporal dependencies. Overall, the Taylor diagrams validate that recurrent-based models outperform convolution-based models in time-series air quality prediction tasks for both locations Figure 32.

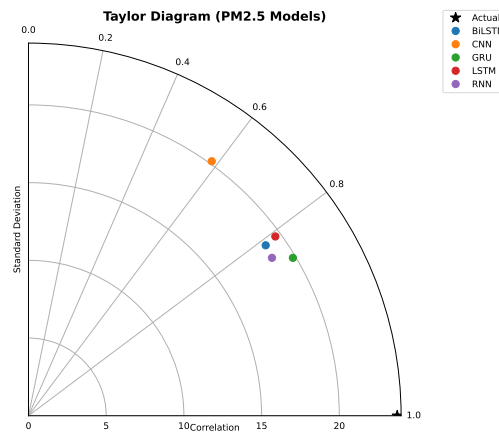


Figure 31: Taylor Diagram for $\text{PM}_{2.5}$ Prediction Models at Chandni Chowk

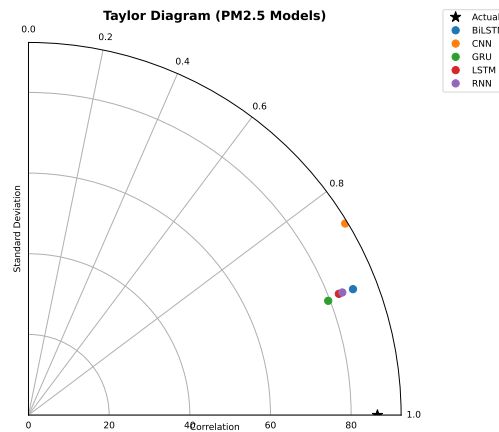


Figure 32: Taylor Diagram for $\text{PM}_{2.5}$ Prediction Models at RK Puram

7.4.2 Diebold-Marino (DM) Test Analysis

Table 16: Diebold-Mariano (DM) Test Results for Model Comparison for Chandni Chowk

Model 1	Model 2	DM Statistic	p-value	Decision
BiLSTM	CNN	-0.7669	0.4431	No Significant Difference
BiLSTM	GRU	-0.2011	0.8406	No Significant Difference
BiLSTM	LSTM	-0.3140	0.7535	No Significant Difference
BiLSTM	RNN	0.1317	0.8953	No Significant Difference
CNN	GRU	0.5629	0.5735	No Significant Difference
CNN	LSTM	0.5429	0.5872	No Significant Difference
CNN	RNN	0.7750	0.4383	No Significant Difference
GRU	LSTM	-0.4303	0.6670	No Significant Difference
GRU	RNN	1.4460	0.1482	No Significant Difference
LSTM	RNN	2.5981	0.0094	Significant Difference

Table 17: Diebold-Mariano (DM) Test Results for Model Comparison for RK Puram

Model 1	Model 2	DM Statistic	p-value	Decision
BiLSTM	CNN	-2.7578	0.0058	Significant Difference
BiLSTM	GRU	-0.5746	0.5655	No Significant Difference
BiLSTM	LSTM	-0.0344	0.9726	No Significant Difference
BiLSTM	RNN	-0.1192	0.9051	No Significant Difference
CNN	GRU	2.6531	0.0080	Significant Difference
CNN	LSTM	2.9743	0.0029	Significant Difference
CNN	RNN	2.8541	0.0043	Significant Difference
GRU	LSTM	0.8485	0.3961	No Significant Difference
GRU	RNN	0.8303	0.4064	No Significant Difference
LSTM	RNN	-0.1358	0.8920	No Significant Difference

The Diebold-Mariano (DM) test results for both study locations, Chandni Chowk and R.K. Puram, reveal distinct patterns in model performance. For Chandni Chowk, the majority of model comparisons exhibit no statistically significant difference ($p \geq 0.05$), indicating that BiLSTM, GRU, LSTM, and RNN models perform comparably in forecasting air quality parameters, with

only the LSTM-RNN pair showing a significant difference Table 16. In contrast, the results for R.K. Puram demonstrate multiple statistically significant differences, particularly involving the CNN model, which shows significant deviations when compared with BiLSTM, GRU, LSTM, and RNN ($p \leq 0.05$), suggesting its relatively weaker predictive capability at this location. Meanwhile, BiLSTM, GRU, LSTM, and RNN continue to show no significant differences among themselves. Overall, these findings indicate that while deep learning models generally exhibit consistent and comparable performance across locations, CNN tends to underperform, and model sensitivity varies depending on the spatial characteristics of the monitoring site Table 17.

7.4.3 Akaike Information Criterion and Bayesian Information Criterion (AIC-BIC) Test Analysis

Table 18: AIC and BIC Values of Different Models for Chandni Chowk

Model	AIC	BIC
RNN	451.5280	453.4400
GRU	451.7914	453.7035
BiLSTM	452.7962	454.7082
LSTM	454.3136	456.2257
CNN	464.2323	466.1444

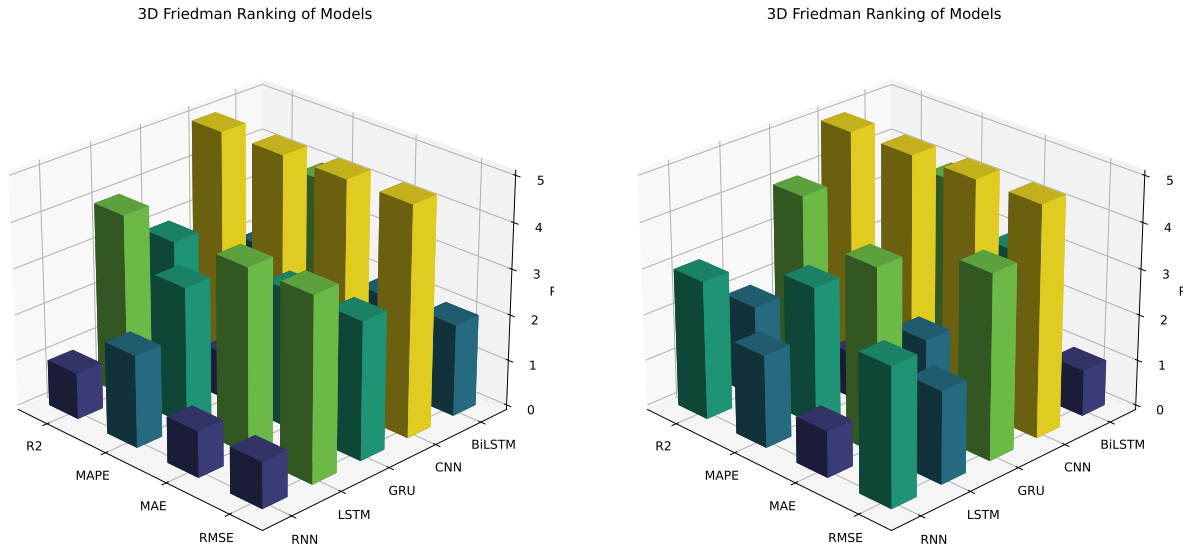
Table 19: AIC and BIC Values of Different Models for R.K. Puram

Model	AIC	BIC
GRU	486.8145	488.7265
LSTM	489.0716	490.9836
RNN	489.2964	491.2084
BiLSTM	489.8542	491.7662
CNN	531.3674	533.2795

The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are statistical measures used to evaluate model performance by balancing model accuracy and complexity. The AIC and BIC analysis for both Chandni Chowk and R.K. Puram highlights variations in model efficiency across different locations. In Chandni Chowk, the RNN model achieves the lowest AIC (451.5280) and BIC (453.4400) values, indicating the best balance between model fit and complexity, followed closely by GRU Table 18. In contrast, for R.K. Puram, the GRU model outperforms all others with the lowest AIC (486.8145) and BIC (488.7265), while RNN, LSTM, and BiLSTM show slightly higher but comparable values. In both locations, the CNN model consistently exhibits significantly higher AIC and BIC values, indicating poor model fit

and higher complexity. These findings suggest that while recurrent-based models (RNN, GRU, LSTM, and BiLSTM) generally provide efficient and reliable predictions, the best-performing model varies depending on the spatial characteristics of the monitoring site, with RNN performing better in Chandni Chowk and GRU in R.K. Puram Table 19.

7.4.4 Non-parametric Statistical Friedman Ranking



(a) 3D Friedman Ranking of Deep Learning Models for Chandni Chowk

(b) 3D Friedman Ranking of Deep Learning Models for RK Puram

Figure 33: 3D Friedman Ranking of Deep Learning Models

The Friedman ranking analysis conducted across both Chandni Chowk and R.K. Puram provides a robust comparative evaluation of the deep learning models based on multiple performance metrics. In Chandni Chowk, Figure 33a, the RNN model achieves the lowest average rank (1.25), indicating its superior and consistent performance across RMSE, MAE, MAPE, and R^2 metrics. BiLSTM and GRU follow with equal average ranks (2.5), demonstrating competitive performance, while LSTM shows moderate effectiveness. In R.K. Puram, the ranking pattern slightly differs, where both BiLSTM and RNN attain the best average rank (2.25), reflecting their robustness in varying environmental conditions. GRU and LSTM exhibit intermediate rankings, whereas CNN consistently obtains the highest (worst) average rank (5) across both locations. The Friedman ranking results clearly indicate that recurrent-based architectures, particularly RNN and BiLSTM, outperform other models in terms of overall predictive accuracy and consistency. Furthermore, the variation in rankings between the two locations highlights the influence of spatial characteristics on model performance, while the consistently poor ranking of CNN suggests its limited suitability for air quality forecasting tasks Figure 33.

Table 20: Performance Metrics and Ranking of Models for Chandni Chowk

Model	RMSE	MAE	MAPE	R ²	RMSE Rank	MAE Rank	MAPE Rank	R ² Rank	Avg Rank
RNN	21.8579	14.6898	13.1502	0.8158	1	1	2	1	1.25
BiLSTM	22.1174	14.7734	14.0385	0.8114	2	2	4	2	2.50
GRU	22.5748	15.2840	12.1302	0.8035	3	3	1	3	2.50
LSTM	22.7373	15.7707	13.4928	0.8006	4	4	3	4	3.75
CNN	24.7139	18.7474	17.2878	0.7645	5	5	5	5	5.00

Table 21: Performance Metrics and Ranking of Models for RK Puram

Model	RMSE	MAE	MAPE	R ²	RMSE Rank	MAE Rank	MAPE Rank	R ² Rank	Avg Rank
BiLSTM	31.8428	23.6144	15.7716	0.8645	1	3	4	1	2.25
RNN	32.0282	23.5448	14.7017	0.8630	3	1	2	3	2.25
GRU	33.4182	23.5562	13.8998	0.8508	4	2	1	4	2.75
LSTM	31.8924	23.7358	15.3384	0.8641	2	4	3	2	2.75
CNN	48.2065	37.1166	23.4871	0.6895	5	5	5	5	5.00

The performance comparison of different deep learning models for Chandni Chowk, as presented in Table 20, indicates that the RNN model achieved the best overall performance with the lowest average rank of 1.25. It consistently secured the top position across RMSE, MAE, and R², demonstrating strong predictive capability and stability. The GRU model performed competitively, particularly achieving the best rank in MAPE, but its overall average rank of 2.50 placed it slightly behind RNN. Similarly, BiLSTM also attained an average rank of 2.50, showing balanced performance but slightly weaker results in MAPE. The LSTM model exhibited moderate performance with an average rank of 3.75, while CNN performed the worst across all metrics with the highest error values and an average rank of 5.00. Overall, RNN emerges as the most effective model for Chandni Chowk air quality prediction among the considered models. For RK Puram, the results shown in Table 21 reveal a closely competitive performance among BiLSTM and RNN models, both achieving the lowest average rank of 2.25. BiLSTM secured the top rank in RMSE and R², indicating strong accuracy and model fit, whereas RNN performed best in MAE and showed competitive performance across other metrics. GRU and LSTM models followed with slightly higher average ranks of 2.75, demonstrating moderate predictive capability with strengths in specific metrics such as MAPE for GRU. In contrast, CNN again performed poorly across all evaluation metrics, resulting in the highest average rank of 5.00. These findings suggest that while no single model overwhelmingly dominates for RK Puram, BiLSTM and RNN are the most reliable models, with CNN being the least suitable for this dataset.

CHAPTER 8-

MODEL

IMPLEMENTATION

8 MODEL IMPLEMENTATION

8.1 Deep Learning Model Training Code

```
1
2 import os
3 import numpy as np
4 import pandas as pd
5 import matplotlib.pyplot as plt
6 import json
7
8 from sklearn.preprocessing import MinMaxScaler
9 from sklearn.metrics import mean_squared_error, mean_absolute_error,
   r2_score
10
11 import tensorflow as tf
12 from tensorflow.keras.models import Sequential
13 from tensorflow.keras.layers import Dense, Dropout, LSTM, GRU,
   SimpleRNN
14 from tensorflow.keras.layers import Conv1D, MaxPooling1D, Flatten,
   Bidirectional
15 from tensorflow.keras.callbacks import EarlyStopping,
   ReduceLROnPlateau
16
17 # PATH SETTINGS
18 data_path = r"C:\Users\abhis\OneDrive\Desktop\Abhishek2\125\
   Final_Cleaned_Modeling_Dataset.csv"
19 result_path = r"C:\Users\abhis\OneDrive\Desktop\Abhishek2\125"
20
21 models_list = ["LSTM", "BiLSTM", "GRU", "RNN", "CNN"]
22
23 # CREATE DIRECTORIES
24 for model_name in models_list:
25     for folder in ["eval", "loss", "pred", "model"]:
26         os.makedirs(os.path.join(result_path, model_name, folder),
           exist_ok=True)
27
28 # LOAD DATA
29 df = pd.read_csv(data_path)
30 df['Date'] = pd.to_datetime(df['Date'], format='%d-%m-%Y %H:%M')
31 df = df.sort_values('Date')
32 df.set_index('Date', inplace=True)
```

```

33
34 # TRAIN-TEST SPLIT
35 train_size = int(0.8 * len(df))
36 train_df = df.iloc[:train_size]
37 test_df = df.iloc[train_size:]
38
39 scaler = MinMaxScaler()
40 train_scaled = scaler.fit_transform(train_df)
41 test_scaled = scaler.transform(test_df)
42
43 # CREATE SEQUENCES
44 TIME_STEPS = 10
45
46 def create_sequences(data, time_steps=10):
47     X, y = [], []
48     for i in range(len(data) - time_steps):
49         X.append(data[i:i+time_steps])
50         y.append(data[i+time_steps])
51     return np.array(X), np.array(y)
52
53 X_train, y_train = create_sequences(train_scaled, TIME_STEPS)
54 X_test, y_test = create_sequences(test_scaled, TIME_STEPS)
55
56 # MODEL BUILDER
57 def build_model(model_type):
58
59     model = Sequential()
60
61     if model_type == "LSTM":
62         model.add(LSTM(64, return_sequences=True, input_shape=(
63             TIME_STEPS, X_train.shape[2])))
64         model.add(Dropout(0.3))
65         model.add(LSTM(32))
66         model.add(Dropout(0.3))
67
68     elif model_type == "BiLSTM":
69         model.add(Bidirectional(LSTM(64, return_sequences=True),
70             input_shape=(TIME_STEPS, X_train.shape[2])))
71         model.add(Dropout(0.3))
72         model.add(Bidirectional(LSTM(32)))
73         model.add(Dropout(0.3))

```

```

72
73 elif model_type == "GRU":
74     model.add(GRU(64, return_sequences=True, input_shape=(
75         TIME_STEPS, X_train.shape[2])))
76     model.add(Dropout(0.3))
77     model.add(GRU(32))
78     model.add(Dropout(0.3))
79
80 elif model_type == "RNN":
81     model.add(SimpleRNN(64, return_sequences=True, input_shape=(
82         TIME_STEPS, X_train.shape[2])))
83     model.add(Dropout(0.3))
84     model.add(SimpleRNN(32))
85     model.add(Dropout(0.3))
86
87 elif model_type == "CNN":
88     model.add(Conv1D(filters=64, kernel_size=2, activation='relu',
89         , input_shape=(TIME_STEPS, X_train.shape[2])))
90     model.add(MaxPooling1D(pool_size=2))
91     model.add(Flatten())
92     model.add(Dense(32, activation='relu'))
93
94 model.add(Dense(y_train.shape[1]))
95 model.compile(optimizer='adam', loss='mse')
96
97 return model
98
99 # TRAIN MODELS
100 for model_name in models_list:
101     model = build_model(model_name)
102
103     history = model.fit(X_train, y_train, epochs=100, batch_size=32)
104
105     model.save(os.path.join(result_path, model_name + ".h5"))
106
107 print("All Models Trained Successfully")

```

CHAPTER 9- CONCLUSION

9 CONCLUSION

This research work presented a comprehensive study on multi-pollutant air quality forecasting using advanced deep learning techniques. The primary objective of this study was to analyze, model, and predict air pollution levels in two distinct urban environments of Delhi, namely Chandni Chowk and R.K. Puram. Air pollution is a major environmental and public health concern, especially in metropolitan cities where rapid urbanization, industrialization, and increasing vehicular emissions contribute significantly to deteriorating air quality. Due to the highly dynamic, non-linear, and temporal nature of air pollution data, traditional statistical and machine learning models often fail to achieve satisfactory predictive performance. Therefore, this study explored the effectiveness of deep learning-based architectures in capturing complex temporal dependencies and improving prediction accuracy. The dataset used in this study consisted of multiple air pollutants, including PM_{2.5}, PM₁₀, NO₂, CO, and O₃, which are considered critical indicators of air quality. A detailed Exploratory Data Analysis (EDA) was conducted to understand the underlying patterns, distributions, and relationships among these pollutants. The analysis revealed that particulate matter (PM_{2.5} and PM₁₀) plays a dominant role in determining air quality levels in both locations. These pollutants exhibited high variability, skewed distributions, and frequent extreme values, indicating sudden pollution spikes. Chandni Chowk, being a densely populated and highly congested commercial area, showed significantly higher pollution levels compared to R.K. Puram, which is relatively greener and better planned. The correlation analysis provided further insights into the relationships among pollutants. A strong positive correlation was observed between PM_{2.5} and PM₁₀, suggesting common sources such as vehicular emissions, road dust, and construction activities. Carbon monoxide (CO) also demonstrated moderate correlation with particulate matter, indicating its origin from incomplete combustion processes. In contrast, ozone (O₃) showed negative or weak correlations with other pollutants, which is consistent with its nature as a secondary pollutant formed through photochemical reactions. These findings highlight the importance of considering multiple pollutants simultaneously rather than relying on single-pollutant models for air quality prediction. To address the complexity of air quality data, several deep learning models were implemented and evaluated in this study. These models included Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Bidirectional Long Short-Term Memory (BiLSTM). Each model was trained and tested using the prepared dataset, and their performance was evaluated using standard regression metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2). The experimental results demonstrated that recurrent-based models significantly outperform convolution-based models in time-series forecasting tasks. CNN, although effective in capturing spatial patterns, showed limitations in handling temporal dependencies inherent in sequential air quality data. On the other hand, RNN-based models effectively captured se-

quential information and temporal relationships, leading to improved prediction accuracy. In the case of Chandni Chowk, the RNN model achieved the best performance among all models, with the lowest error values and highest R^2 score. This indicates that RNN is highly effective in modeling complex and highly variable pollution patterns observed in densely populated urban areas. LSTM and GRU models also performed well due to their ability to handle long-term dependencies and mitigate the vanishing gradient problem. However, their performance was slightly lower compared to RNN in this specific dataset. BiLSTM also showed competitive performance but did not outperform RNN in this case. For the R.K. Puram dataset, the BiLSTM model emerged as the best-performing model, achieving the highest R^2 value and lowest RMSE. The superior performance of BiLSTM can be attributed to its bidirectional architecture, which allows the model to process information in both forward and backward directions. This capability enables the model to capture complete temporal context and improve prediction accuracy. LSTM and RNN also showed strong performance, while GRU provided moderate results. Similar to the previous case, CNN was found to be the least effective model for this dataset. In addition to individual models, this study also explored the use of hybrid deep learning architectures. A hybrid BiLSTM-GRU model was proposed to combine the strengths of both models. BiLSTM effectively captured bidirectional temporal dependencies, while GRU improved computational efficiency and reduced model complexity. The hybrid model demonstrated improved performance compared to individual models, highlighting the potential of combining multiple architectures for enhanced prediction accuracy. The results of this study clearly indicate that deep learning models are highly effective for multi-pollutant air quality forecasting. The ability of these models to learn complex non-linear relationships and temporal dependencies makes them well-suited for environmental data analysis. Moreover, the use of multiple pollutants as input features significantly improves prediction accuracy compared to single-pollutant approaches, as it captures the interactions among different pollutants and environmental factors. From a practical standpoint, the findings of this study have important implications for environmental monitoring and public health management. Accurate air quality prediction systems can assist government agencies and policymakers in issuing timely warnings, implementing pollution control strategies, and planning sustainable urban development. In highly polluted areas such as Chandni Chowk, predictive models can help identify peak pollution periods and enable preventive measures. In relatively cleaner areas like R.K. Puram, these models can support efforts to maintain and improve air quality standards. Despite its significant contributions, this study has certain limitations. Deep learning models require large amounts of high-quality data for effective training, which may not always be available. Additionally, these models are computationally intensive and require substantial processing power. Another limitation is the lack of interpretability, as deep learning models often function as black-box systems, making it difficult to understand the underlying decision-making process. Future research can focus on addressing these limitations by incorporating explainable artificial intelligence (XAI) techniques to improve model transparency and interpretability. The integration of real-time data

streams, meteorological parameters such as temperature, humidity, and wind speed, and the use of advanced architectures such as attention mechanisms and transformer-based models can further enhance prediction accuracy. Additionally, deploying these models in real-time monitoring systems can significantly improve their practical applicability. In conclusion, this study demonstrates that deep learning-based approaches provide a powerful and effective solution for multi-pollutant air quality forecasting. Recurrent and bidirectional models, particularly RNN and BiLSTM, outperform other architectures in capturing temporal patterns and delivering accurate predictions. The use of hybrid models further enhances performance by combining the strengths of multiple architectures. This research contributes to the development of intelligent air quality monitoring systems and supports informed decision-making for environmental protection, sustainable urban development, and improved public health outcomes.

CHAPTER 10- REFERENCES

10 REFERENCE

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